Discrete Structures & Optimization

Mathematical Logic



Discrete Structures and Optimization

Mathematical Logic:

Propositional and Predicate Logic, Propositional Equivalences, Normal Forms, Predicates and Quantifiers, Nested Quantifiers, Rules of Inference.

Sets and Relations:

Set Operations, Representation and Properties of Relations, Equivalence Relations, Partially Ordering.

Counting, Mathematical Induction and Discrete Probability:

Basics of Counting, Pigeonhole Principle, Permutations and Combinations, Inclusion- Exclusion Principle, Mathematical Induction, Probability, Bayes' Theorem .

Group Theory:

Groups, Subgroups, Semi Groups, Product and Quotients of Algebraic Structures, Isomorphism, Homomorphism, Automorphism, Rings, Integral Domains, Fields, Applications of Group Theory.

Graph Theory:

Simple Graph, Multigraph, Weighted Graph, Paths and Circuits, Shortest Paths in Weighted Graphs, Eulerian Paths and Circuits, Hamiltonian Paths and Circuits, Planner graph, Graph Coloring, Bipartite Graphs, Trees and Rooted Trees, Prefix Codes, Tree Traversals, Spanning Trees and Cut-Sets.

Boolean Algebra:

Boolean Functions and its Representation, Simplifications of Boolean Functions.

Optimization:

Linear Programming - Mathematical Model, Graphical Solution, Simplex and Dual Simplex Method, Sensitive Analysis; Integer Programming, Transportation and Assignment Models, PERT-CPM: Diagram Representation, Critical Path Calculations, Resource Levelling, Cost Consideration in Project Scheduling.

Introduction to Discrete Mathematics

Why study Discrete Mathematics?

- To develop mathematical thinking.
- To improve problem solving ability.
- Foundation for other compute science subjects.
- To solve problems such as:
  - Sorting the list of integers,
  - Finding shortest path between places,
  - Drawing graphs without lifting the pen and without repeating edges,
  - Message Encryption, etc…

Introduction to Discrete Mathematics

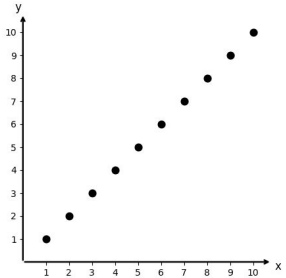
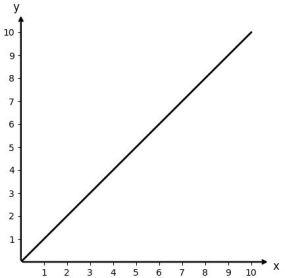
What is Discrete Mathematics?

- The study of discrete objects.
- Discrete means distinct or not connected.
- A branch of mathematics dealing with

"Discrete but not continuous"

Introduction to Discrete Mathematics

Discrete vs Continuous

Discrete Objects	Continuous Objects
Natural numbers, N = 1, 2, 3, … Consider the graph for $y=x$ where x,y belongs to N, gaps are present between points. 	Real numbers, R between two numbers 0 and 1, there are infinite number of continuous points, 0.0001, 0.00001,... Consider the graph for $y=x$ where x,y belongs to R, continuous line is plotted with no gaps in between. 
Digital clock: No continuous time, sharp transition from one time to another time.	Analog clock: Hour, minute and second hands sweep smoothly over time.

Mathematical Logic

- Propositional and Predicate Logic
- Propositional Equivalences
- Normal Forms
- Predicates and Quantifiers
- Nested Quantifiers
- Rules of Inference

Introduction to Propositional Logic

What is Logic?

- The science of reasoning.
- To understand & reason about different mathematical statements.
- Rules of logic enable to
  - Think about mathematical statements,
  - Prove or disprove mathematical statements precisely.
  - Reason about mathematical statements.
- "For every positive integer, the sum of positive integers not exceeding n is n(n+1)/2".

  Either it is a valid mathematical argument or it is an invalid argument.

Introduction to Propositional Logic

Purpose of Logic?

- To construct valid arguments or proofs
- Once we prove a mathematical statement is true, then we call it a Theorem.
- **Famous Knights and Knaves Puzzle:**

  In an Island there are two kinds of inhabitants, Knights, who always tell the truth, and their opposites, Knaves, who always lie. You encounter two people Alice and Bob. Who are Alice and Bob if Alice says "Bob is a Knight" and Bob says "The two of us are opposite types"?

- These types of problems can be easily solved by an area of logic called **propositional logic**.

Propositional Logic

What is proposition?

- A declarative sentence that has a definite truth value.
- A sentence that is declaring a fact or stating an argument which can be either TRUE or FALSE but cannot be both.
- *Atomic Propositions:* A simple statement without logical connectives that cannot be broken down further.
- *Compound Propositions:* A statement formed by combining atomic propositions using logical connectives like AND, OR, or NOT.
- *Non-Propositions:* Questions, commands, and opinions with no definite truth value or may vary depending on context:
- Delhi is the capital of India. ✅
- Water froze this morning. ✅
- 1 + 1 = 2 ✅
- What time is it? ❌
- x + 1 = 2 ❌
- Send us your resume before 11 PM. ❌
- I request you to please allow me a day off. ❌
- Fetch my umbrella! ❌

Which of the following is/are proposition(s). Also if you find the proposition, state whether it is True or False:

1. London is in Denmark. ✅ F
2. Do your Homework. ❌
3. India wins the match by 2 runs. ✅ T/F
4. x is an even number. ❌
5. 5 is an odd number. ✅ T
6. Rahul. ❌
7. 5 + 7 = 10 ✅ F
8. The moon is made of cheese. ✅ F
9. The only odd prime number is 2. ✅ F
10. God bless you! ❌

Propositional Logic

Sentential / Statement Logic

- Branch of formal logic that deals with propositions: statements that are either true or false, and the logical relationships between them, using connectives like "and," "or," "not," and "if...then."
- Area of logic that studies ways of joining and/or modifying propositions to form more complicated propositions.
- Studies the logical relationships and properties derived from these combined/altered propositions.
- Logical connectives are used to join two propositions.

  Statement 1: Adam is good in playing football.
  Statement 2: This time he is representing his college at national level.
  Adam is good in playing football and this time he is representing his college at national level.
- Modifying a statement using negatives.

  I enjoy watching television.
  It is not the case that I enjoy watching television.

Propositional Logic

Why compound propositions?

Most mathematical statements are constructed by combining one or more than one propositions.

Propositional Variables

Variables that are used to represent propositions.

Adam is good in playing football and this time he is representing his college at national level.

p = Adam is good in playing football.
q = this time he is representing his college at national level.

$p \wedge q$ means p **and** q

NET December 2018

In mathematical logic, which of the following are statements ?

(i) There will be snow in January

(ii) What is the time now ?

(iii) Today is Sunday

(iv) You must study Discrete Mathematics.

i ➔ (iii) and (iv)

ii ➔ (i) and (ii)

iii ➔ (i) and (iii) ✅

iv ➔ (ii) and (iv)

Logical Operators

Logical connectives combine simple propositions into compound ones using a small set of connectives. Its truth value depends on the truth values of its parts across every possible scenario.
Truth table: compilation of all possible scenarios in tabular format.

1. Negation
2. Conjunction
3. Disjunction
4. Exclusive OR
5. Implication
6. Biconditional

p	¬p
T	F
F	T

Negation

- Let p be a proposition.
- **¬p** is called negation of p which simply states that "It is **not** the case that p" .
- if p is true then ¬p is false.
- if p is false then ¬p is true.
- p: "Adam and Eve lived together for many years."
- ¬p: "It is not the case that Adam and Eve lived together for many years."

or

- ¬p: "Adam and Eve haven't lived together for many years."

Logical Operators

Conjunction

- Let p and q be two propositions.
- When both p and q are true, then **p ∧ q** is true.
- 'and' or 'but'
- 12 is divisible by 3 **and** 3 is a prime number. (T)

p	q	p ∧ q
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction

- When both p and q are false, then **p ∨ q** is false.
- 16 - 4 = 10 **or** 4 is an even number. (T)
- In order to get a job in this multinational company, **experience with C++ or Java** is mandatory.
- ***Inclusive OR***: **Either** p **or** q or **both** (p as well as q).

p	q	p ∨ q
T	T	T
T	F	T
F	T	T
F	F	F

Logical Operators

Exclusive OR

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

- Let p and q be two propositions.
- Exactly one of p and q will be true but both cannot be true.
- Exudes the *both* possibility.
- When you buy a car from XYZ company, you get $2500 cashback or accessories worth $2500.
- p: you get $2500 cashback.
- q: you get accessories worth $2500
- **$p \oplus q$: Either** p **or** q **but not** both.

1. Coffee or Tea comes with dinner. XOR
2. You can pay using US dollars or euros. XOR
3. Dinner for two includes two items from column A or three items from column B. XOR
4. A password must have at least three digits or be at least eight characters long. OR
5. To take discrete mathematics, you must have taken calculus or a course in CS. OR

Logical Operators

p	q	p → q
T	T	T
T	F	F
F	T	T
F	F	T

Implication

- Also known as Conditional statement.
- **p → q:** If p then q.
- p is called the hypothesis or antecedent or premise.
- q is called the conclusion or consequence.
- Does not guarantee anything when the premise p is false.
- There is no way of knowing whether or not the implication is false since p did not happen.
- This situation is similar to the "Innocent until proven Guilty" stance, which means that the implication p→q is considered true until proven false.

Logical Operators

Implication

Example 1: "If you try hard for your exam, then you will succeed".

p = you tried hard for your exam. q = you succeed

Case 1: "You tried hard for your exam" and "you succeed"

p = True q = True p → q is True

Case 2: "You tried hard for your exam" but "you failed"

p = True q = False p → q is False

Case 3: "You haven't tried hard for your exam" and "You succeeded"

p = False q = True p → q is True

Case 4: "You haven't tried hard for your exam" and "You failed"

p = False q = False p → q is True.

If p itself is not satisfied then compound proposition cannot be False.

Not False means True.

If p is false then it doesn't matter what will be the truth value of q, p → q is always true.

1. If 1 + 1 = 3, then dogs can fly. ✅
2. If 1 + 1 = 2, then dogs can fly. ❌
3. If monkeys can fly, then 1 + 1 =3 ✅
4. If 1 + 1 = 2, then 2 + 2 = 5. ❌
5. If Delhi is the capital of India then Beijing is the capital of China. ✅

- "If you have connection with seniors, then you will get promoted."
- "If you get 100% marks on the final exam, then you will be awarded a Trophy."

Logical Operators

Representations of Implication

"if p then q"
"p implies q"
"q when p"
"q whenever p"
"q follows from p"

"p only if q"
"q is necessary for p"
"p is sufficient for q"
"q unless ¬p"

How "if p then q" and "p only if q" can be same?

Example: "I will stay at home only if I'm sick."

p = "I will stay at home" q = "I'm sick"

Above statement is of the form: p only if q, becoming sick is the necessary condition that will make you stay at home. This means "if you're not sick then, you cannot stay at home at any cost."

In order to falsify the above statement, q must be FALSE and p must be TRUE

i.e. you are not sick and you still stay at home.

Proof idea: truth value of p and q must be same in order to falsify the statement.

if p then q: "If I'll stay at home then I'm sick"

The only way to falsify the above statement is by making p TRUE and q FALSE. Therefore, p only if q is equivalent to if p then q

HENCE PROVED

Logical Operators

Representations of Implication

"if p then q"
"p implies q"
"q when p"
"q whenever p"
"q follows from p"

"p only if q"
"q is necessary for p"
"p is sufficient for q"
"q unless ¬p"

Why "p only if q" is not equivalent to "if q then p"?

Example: "I'll stay at home only if I'm sick."

p = I'll stay at home q = I'm sick

Proof: p only if q is false when p is TRUE and q is FALSE

If q then p: "If I'm sick then I'll stay at home."

FALSE TRUE = TRUE

As when p is TRUE and q is FALSE, "p only if q" is false but with same truth values of p and q, "if q then p" is true.

Therefore, "p only if q" is not equivalent to "if q then p".

HENCE PROVED

If p is true then q must be true for the compound proposition to be true.
If q is false then p must be false for the compound proposition to be true.

Logical Operators

Representations of Implication

Necessary Condition: This condition must be present for the outcome to happen, but its presence does not ensure the outcome. It is the "then" part of an if-then statement. For example, being a rectangle is a necessary condition for being a square, because you cannot have a square without it being a rectangle.

"if p then q"
"p implies q"
"q when p"
"q whenever p"
"q follows from p"

"p only if q"
"q is necessary for p"
"p is sufficient for q"
"q unless ¬p"

"q is necessary for p"

Example: "Good food is necessary to keep us alive"

According to the statement, if we won't have good food then, we'll soon die.

But is it the only factor to keep us alive? **No.**

Example: My friend died at the age of 16 and he has no shortage of good food. (Even though he is having good food but still he died.)

Therefore, when we say A is necessary for B then falsity of A guarantees the falsity of B but *we cannot guarantee* the *truth* of B from the truth of A.

Similarly, when we say q is necessary for p, then we can only guarantee that when q is false then, p is definitely false.

Logical Operators

Representations of Implication

"if p then q"
"p implies q"
"q when p"
"q whenever p"
"q follows from p"

"p only if q"
"q is necessary for p"
"p is sufficient for q"
"q unless ¬p"

"p is sufficient for q"

Example: "It is sufficient for you to travel by car in order to reach your destination on time."

Definitely, if you travel by car, you'll reach your destination on time. No doubt. But if you won't travel by car, does it mean you'll never reach your destination on time? May be by flight or other means of transport, you'll reach your destination much earlier.

Therefore, when we say that when A is sufficient for B then, truth of A guarantees the truth of B but *we cannot guarantee* the *falsity* of B from the falsity of A.

Similarly, when we say "p is sufficient for q", then we can only guarantee that when p is true then, q is definitely true.

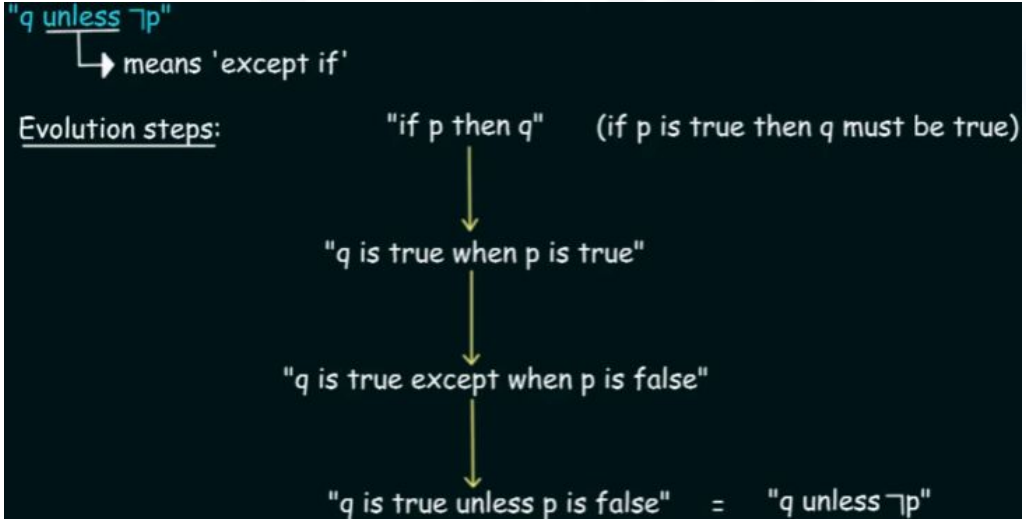
Sufficient Condition: If this condition is met, the outcome is certain. It is the "if" part of an if-then statement. For example, being a square is a sufficient condition for being a rectangle, because every square is a rectangle.

Logical Operators

Semicolon

Representations of Implication

"q unless ¬p"



Why p is not necessary for q?

Proof by contradiction: Let say p is necessary for q.

If p is really necessary for q then if p is false then it is mandatory for q to be false in order to make the whole compound proposition true. But it is not the case. According to the truth table, it is not mandatory for q to be false when p is false. Hence "p is not necessary for q."

Why q is not sufficient for p?

Proof by contradiction: Let say q is sufficient for p.

If q is really sufficient for p then if q is true then it is mandatory for p to be true in order to make the whole compound proposition true. But it is not the case. According to the truth table it is not mandatory for p to be true when q is true. Hence "q is not sufficient for p."

Converse, Contrapositive & Inverse

Implication:	**p → q**	"If it rains today then, I will stay at home."
Converse:	**q → p**	"If I will stay at home, then it rains today."
Contrapositive:	**¬q → ¬p**	"If I will not stay at home, then it does not rain today."
Inverse:	**¬p → ¬q**	"It it does not rain today then, I will not stay at home."

Implication & contrapositive are equivalent, whereas converse and inverse are equivalent.

Neither converse nor inverse is equivalent to implication.

p	q	¬p	¬q	p → q	q → p	¬q → ¬p	¬p → ¬q
T	T	F	F	T	T	T	T
T	F	F	T	F	T	F	T
F	T	T	F	T	F	T	F
F	F	T	T	T	T	T	T

- If you receive A grade in CS then, you will be awarded a scholarship.
- If exercise is good for health then, I will go to the park.

Biconditional Operator (Double Implication)

p ↔ q **p if and only if q** **(p → q) ∧ (q → p)**

True whenever the truth values of p and q are same.

p is necessary and sufficient for q and vice versa

if p then q and conversely (p iff q).

p if and only if q is composed of **p if q and p only if q**

p only if q = if p then q and p if q = if q then p

So, **(p → q) and (q → p) ≡ p ↔ q**

p	q	p ↔ q
T	T	T
T	F	F
F	T	F
F	F	T

p: "You get promoted" q: "You have connections" p ↔ q: "You get promoted if and only if you have connections."

If you read newspaper everyday, you will be informed and conversely. You read the newspaper everyday if and only if you will be informed.

It rains if it is a weekend day, and it is a weekend day if it rains. It rains if and only if it is a weekend day.

The trains run late on exactly those days when I take it. The trains run late if and only if I take it.

For you to get an A in this course, it is necessary and sufficient that you learn how to solve discrete mathematics problems. You get an A in this course if and only if you learn how to solve discrete mathematics problems.

Logical Operators

NET November 2017 Paper 3

"If X, then Y unless Z" is represented by which of the following formulae in propositional logic?

i ➔ $(X \land Y) \rightarrow \neg Z$ **ii ➔** $(X \land \neg Z)$ ✅
Y

iii ➔ $X \rightarrow (Y \land \neg Z)$ **iv ➔** $Y \rightarrow (X \land \neg Z)$

If the statement "p ∧ q" is true, which of the following must also be true?

i ➔ p is true ✅ **ii ➔** q is false **iii ➔** p is false **iv ➔** None of the above

If p ↔ q is true, which of the following must also be true?

i ➔ p and q have the same truth value ✅ **ii ➔** p is true **iii ➔** q is false **iv ➔** p and q have opposite truth values

If p ∨ q is true and ¬p is true, then q must be true.

i ➔ Valid ✅ **ii ➔** Invalid **iii ➔** Valid only if p is true **iv ➔** Can't be determined

Logical Operators

p: "Today is Friday" q: "It is raining today"

¬p: “Today is not friday” ¬q: “It is not raining today”

p ∧ q: "Today is Friday and it is raining today."

p ∨ q: "Today is Friday or it is raining today."

p ⊕ q: "Either today is Friday or it is raining today, but not both."

p → q: "If it is Friday, then it is raining today"

p ↔ q: "It is raining today if and only if it is Friday today"

Consider the following statements:

- **P: Good mobile phones are not cheap.**
- **Q: Cheap mobile phones are not good.**
- **L: P implies Q**
- **M: Q implies P**
- **N: P is equivalent to Q**

Which one of the following about L, M, and N is CORRECT?

(A) Only L is TRUE.

(B) Only M is TRUE.

(C) Only N is TRUE.

(D) L, M, and N are TRUE. ✅

Precedence of Logical Operators

Helps to decide which operator will get evaluated first in a complicated compound proposition.

Operators	Names	Precedence
¬	Negation	1
∧	Conjunction	2
∨	Disjunction	3
→	Implication	4
↔	Biconditional	5

$p \rightarrow q \wedge \neg p$				
p	**q**	**¬p**	**q ∧ ¬p**	**p → q ∧ ¬p**
T	T	F	F	F
T	F	F	F	F
F	T	T	T	T
F	F	T	F	T

$p \rightarrow \neg q \wedge \neg p \leftrightarrow \neg q$						
p	**q**	**¬p**	**¬q**	**¬q ∧ ¬p**	**p → ¬q ∧ ¬p**	**p → ¬q ∧ ¬p ↔ ¬q**
T	T	F	F	F	F	T
T	F	F	T	F	F	F
F	T	T	F	F	T	F
F	F	T	T	T	T	T

Translating English Sentences into Logical Expressions

- Removes ambiguity
- Easy manipulation
- Able to solve puzzles

"You are not allowed to marry if your age is less than 18 years or you have no age proof."

- q: "You are allowed to marry"
- r: "Your age is less than 18 years"
- s: "You have age proof"

Statement follows the structure "A if B" (which means If B then A)

The logical expression: $\mathbf{(r \vee \neg s) \rightarrow \neg q}$

Are these system specifications consistent?

1. "The system is in multiuser state if and only if it is operating normally."
2. "If the system is operating normally, then the kernel is functioning."
3. "The kernel is not functioning or the system is in interrupt mode."
4. "If the system is not in multiuser state, then it is in interrupt mode."
5. "The system is not in interrupt mode

Consistent means assigning truth values to propositional variables in such a way that finally we would be able to make all specifications 'true'.

p = "The system is in multiuser state." q = "The system is operating normally."
r = "The kernel is functioning." s = "The system is in interrupt mode."

$p \leftrightarrow q$ $q \rightarrow r$ $\neg r \vee s$ $\neg p \rightarrow s$
$\neg s$

Always start from the statement which involves least number of propositions

In an Island there are two kinds of inhabitants, Knights, who always tell the truth, and their opposites, Knaves, who always lie. You encounter two people A and B. Determine, if possible, what A and B are if they address you in the ways described:

a. A says "B is a Knight" and B says "The two of us are opposite types".

Let **p = A is knight** and **q = B is knight**

So **¬p = A is knave** and **¬q = B is knave**

Case 1:

Suppose **A is knight**: p is T

Whatever A says will be true: q is T

So B is knight, whatever B says must be true:

$(p \wedge \neg q) \vee (\neg p \wedge q)$ must be T, but became F.

Hence **A is knave**: a contradiction. ❌

Case 2:

Suppose **B is knight**: q is T

Whatever B says will be true:

$(p \wedge \neg q) \vee (\neg p \wedge q)$ will be T, only if p is F

So A is knave, whatever A says will be false: q is F

Hence **B is knave**: a contradiction. ❌

Case 3:

Suppose **A is knave**: p is F

Whatever A says will be false: q is F

So B is knave, whatever B says must be false:

$(p \wedge \neg q) \vee (\neg p \wedge q)$ must be F, and is proved to be F.

Hence **A is knave** ✅

Case 4:

Suppose **B is knave**: q is F

Whatever B says will be false:

$(p \wedge \neg q) \vee (\neg p \wedge q)$ will be F, only if p is F

So A is knave, whatever A says will be false: q is F

Hence **B is knave** ✅

Hence A is a knave and B is a knave.

In an Island there are two kinds of inhabitants, Knights, who always tell the truth, and their opposites, Knaves, who always lie. You encounter two people A and B. Determine, if possible, what A and B are if they address you in the ways described:

b. A says "Atleast one of us is a knave" and B says nothing.

Let **p = A is knight** and **q = B is knight**

So **¬p = A is knave** and **¬q = B is knave**

Case 1:

Suppose **A is knight**: p is T

Whatever A says will be true:

Either A or B or both are knaves:

$(p \wedge \neg q) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q)$ will be T, only if q is F.

So **B is knave** and hence **A is knight** ✅

Case 2:

Suppose **B is knight**: q is T

Whatever B says will be true:

But B said nothing ❌

Case 3:

Suppose **A is knave**: p is F

Whatever A says will be false:

None of them are knaves

But then A cannot be knave

Hence **A is knight: a contradiction** ❌

Case 4:

Suppose **B is knave**: q is F

Whatever B says will be false:

But B said nothing ❌

Hence A is a knight and B is a knave.

Semicolon

In an Island there are two kinds of inhabitants, Knights, who always tell the truth, and their opposites, Knaves, who always lie. You encounter two people A and B. Determine, if possible, what A and B are if they address you in the ways described:

c. A says "The two of us are both knights" and B says "A is a knave".

Let **p = A is knight** and **q = B is knight**

Case 1:
Suppose **A is knight**: p is T
Whatever A says will be true:
So B is also a knight q is T
Whatever B says should be true.
So **A is a knave**: a contradiction ❌

Case 2:
Suppose **B is knight**: q is T
Whatever B says will be true, so A is a knave p is F
Whatever A says will be false.
So **A is a knave** and hence **B is knight** ✅

So **¬p = A is knave** and **¬q = B is knave**

Case 3:
Suppose **A is knave**: p is F
Whatever A says will be false:
As B says A is a knave, **B is a knight**
Hence **A is knave** ✅

Case 4:
Suppose **B is knave**: q is F
Whatever B says will be false: p is T
So A is a knight, whatever A says should be true.
Hence **B is a knight**: a contradiction ❌

Hence A is a knave and B is a knight.

In an Island there are two kinds of inhabitants, Knights, who always tell the truth, and their opposites, Knaves, who always lie. You encounter two people A and B. Determine, if possible, what A and B are if they address you in the ways described:

d. A says "I am a knave or B is a knight" and B says nothing.

Let **p = A is knight** and **q = B is knight**

Case 1:
Suppose **A is knight**: p is T
Whatever A says will be true:
So A is a knave or B is a knight must be true:
¬p ∨ q is T, only if q is T.
So **B is knight**. and hence **A is knight**: ✅

Case 2:
Suppose **B is knight**: q is T
Whatever B says will be true:
But B said nothing ❌

So **¬p = A is knave** and **¬q = B is knave**

Case 3:
Suppose **A is knave**: p is F
Whatever A says will be false:
So A is a knave or B is a knight must be false:
But ¬p ∨ q is always T, as p is F
Hence **A is knight**: a contradiction ❌

Case 4:
Suppose **B is knave**: q is F
Whatever B says will be false:
But B said nothing ❌

Hence A is a knight and B is a knight.

In an Island there are two kinds of inhabitants, Knights, who always tell the truth, and their opposites, Knaves, who always lie. You encounter two people A and B. Determine, if possible, what A and B are if they address you in the ways described:

e. Both A and B say "I am a knight".

Let **p = A is knight** and **q = B is knight**

Case 1:
Suppose **A is knight**: p is T
Whatever A says will be true:
So **A is a knight**

Case 2:
Suppose **B is knight**: q is T
Whatever B says will be true:
So **B is a knight**

So **¬p = A is knave** and **¬q = B is knave**

Case 3:
Suppose **A is knave**: p is F
Whatever A says will be false:
So **A is a knave**

Case 4:
Suppose **B is knave**: q is F
Whatever B says will be false:
So B is a knave

It is impossible to determine the identities of A and B based solely on these statements.

Semicolon

A detective has interviewed four witnesses to a crime. From the stories of the witnesses the detective has concluded that if the butler is telling the truth then so is the cook; the cook and the gardener cannot both be telling the truth; the gardener and the handyman are not both lying; and if the handyman is telling the truth then the cook is lying. For each of the four witnesses, can the detective determine whether that person is telling the truth or lying?

Based on the logical constraints provided, the detective can determine the status of the **butler** and the **cook**, but **cannot** definitively determine the status of the **gardener** and the **handyman**.

Logical Deduction

Let the variables B, C, G, and H represent the truthfulness of the Butler, Cook, Gardener, and Handyman, respectively (where True means telling the truth and False means lying).

The premises are:

1. $B \rightarrow C$ (If Butler is true, Cook is true)
2. $\neg(C \wedge G)$ (Cook and Gardener cannot both be true)
3. $G \vee H$ (Gardener and Handyman are not both lying; at least one is true)
4. $H \rightarrow \neg C$ (If Handyman is true, Cook is lying)

Step 1: Determine the Cook's status (C) Assume the Cook is telling the truth ($C = \text{True}$):

- From premise (4), if C is True, then H must be False (by contrapositive: $C \rightarrow \neg H$).
- From premise (2), if C is True, then G must be False.
- Now we have $G = \text{False}$ and $H = \text{False}$.
- This contradicts premise (3), which states that G and H cannot both be False.
- **Conclusion:** The assumption is false. The **Cook is lying** ($C = \text{False}$).

Step 2: Determine the Butler's status (B)

- From premise (1), we have $B \rightarrow C$.
- Since we know C is False, B must also be False (Modus Tollens).
- **Conclusion:** The **Butler is lying** ($B = \text{False}$).

Step 3: Analyze the Gardener (G) and Handyman (H) We know $C = \text{False}$. Let's test the remaining possibilities for G and H under premises (3) and (4):

- Premise (4) ($H \rightarrow \neg C$) is satisfied regardless of H because $\neg C$ is True.
- Premise (2) ($\neg(C \wedge G)$) is satisfied regardless of G because C is False.
- We are left with premise (3): $G \vee H$ must be True.

This allows for multiple valid scenarios:

1. **Gardener True, Handyman True:** Consistent with all rules.
2. **Gardener True, Handyman False:** Consistent with all rules.
3. **Gardener False, Handyman True:** Consistent with all rules.

Since multiple scenarios exist where the truth values of G and H differ, their specific statuses cannot be uniquely determined.

Final Conclusion

- **Butler: Lying**
- **Cook: Lying**
- **Gardener: Undetermined** (Could be True or False)
- **Handyman: Undetermined** (Could be True or False)

In a room there are only two types of people, namely Type 1 and Type 2. Type 1 people always tell the truth and Type 2 people always lie. You give a fair coin to a person in that room, without knowing which type he is from and tell him to toss it and hide the result from you till you ask for it. Upon asking, the person replies the following. "The Result of the toss is head if and only if I am saying the truth." Which of the following options is correct? (a) The result is head. (b) The result is tail. (c) If the person is of Type 2, then the result is tail. (d) If the person is of Type 1, then the result is tail. [GATE 2015 (Set - 3)]

The correct option is **(a) The result is head.**

Logical Deduction

Let H represent the proposition "The result is Head" and T represent "The person is telling the truth" (i.e., the person is Type 1). The statement made by the person is: $H \leftrightarrow T$ ("Head if and only if I am telling the truth").

We analyze the two possible cases for the person's type:

Case 1: The person is Type 1 (Truth-teller)

- Since Type 1 always tells the truth, T is **True**.
- Therefore, the statement ($H \leftrightarrow T$) must be **True**.
- Substituting $T = \text{True}$: $H \leftrightarrow \text{True}$ is True.
- This implies H **is True**.
- **Result:** The coin is Head.

Case 2: The person is Type 2 (Liar)

- Since Type 2 always lies, T is **False**.
- Therefore, the statement ($H \leftrightarrow T$) must be **False**.
- Substituting $T = \text{False}$: $H \leftrightarrow \text{False}$ is False.
- For a biconditional ($H \leftrightarrow \text{False}$) to be False, H must be the opposite of False.
- This implies H **is True**.
- **Result:** The coin is Head.

Conclusion

In both possible scenarios, the result of the toss is **Head**. Therefore, the detective can definitively conclude that the result is Head, regardless of whether the person is Type 1 or Type 2.

Tautology

Compound proposition which is always **TRUE**.

Eg: $p \vee \neg p$

Contradiction

Compound proposition which is always **FALSE**.

Eg: $p \wedge \neg p$

Contingency

Compound proposition which is **sometimes TRUE and sometimes FALSE**.

Eg: $p \wedge q$

Valid

A compound proposition is valid when it is a **tautology**.

Invalid

A compound proposition is invalid when it is **either a contradiction or contingency**.

Satisfiability

A compound proposition is satisfiable if there is **at least one TRUE** result in its truth table.

Eg: $p \wedge q$ is satisfiable.

A **tautology** is always satisfiable but satisfiable is not always tautology.

Unsatisfiability

A compound proposition is unsatisfiable if there is **not even a single TRUE** result in its truth table.

A **contradiction** is always unsatisfiable.

Tautology	Contradiction	Contingency
T	F	T/F
Satisfiable	Unsatisfiable	Satisfiable
Valid	Invalid	Invalid

Logical Equivalences

The compound propositions p and q are said to be logically equivalent **if p↔q is a tautology.**
Denoted by ≡ or ⇔

Identity Laws	$p \vee F \equiv p$	$p \wedge T \equiv p$
Domination Laws	$p \vee T \equiv T$	$p \wedge F \equiv F$
Idempotent Laws	$p \vee p \equiv p$	$p \wedge p \equiv p$
Double Negation Law	$\neg(\neg p) \equiv p$	
Commutative Laws	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
Associative Laws	$(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
Distributive Laws	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
De Morgan's Laws	$\neg(p \wedge q) \equiv \neg p \vee \neg q$	$\neg(p \vee q) \equiv \neg p \wedge \neg q$
Absorption Laws	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
Negation Laws	$p \vee \neg p \equiv T$	$p \wedge \neg p \equiv F$

1. Algebraic Proof

Law 1: $p \vee (p \wedge q) \equiv p$

- **Step 1:** Apply the **Identity Law** ($p \equiv p \wedge \text{True}$): $(p \wedge \text{True}) \vee (p \wedge q)$
- **Step 2:** Apply the **Distributive Law** (factor out p): $p \wedge (\text{True} \vee q)$
- **Step 3:** Apply the **Domination/Null Law** ($\text{True} \vee q \equiv \text{True}$): $p \wedge \text{True}$
- **Step 4:** Apply the **Identity Law** again: p

Law 2: $p \wedge (p \vee q) \equiv p$

- **Step 1:** Apply the **Identity Law** ($p \equiv p \vee \text{False}$): $(p \vee \text{False}) \wedge (p \vee q)$
- **Step 2:** Apply the **Distributive Law** (factor out p): $p \vee (\text{False} \wedge q)$
- **Step 3:** Apply the **Domination/Null Law** ($\text{False} \wedge q \equiv \text{False}$): $p \vee \text{False}$
- **Step 4:** Apply the **Identity Law** again: p

Logical Equivalences

Semicolon

Conditionals
$p \rightarrow q \equiv \neg p \lor q$
$p \rightarrow q \equiv \neg q \rightarrow \neg p$
$p \lor q \equiv \neg p \rightarrow q$
$p \land q \equiv \neg(q \rightarrow \neg p)$
$\neg(p \rightarrow q) \equiv p \land \neg q$
$(p \rightarrow q) \land (p \rightarrow r) \equiv p \rightarrow (q \land r)$
$(p \rightarrow q) \lor (p \rightarrow r) \equiv p \rightarrow (q \lor r)$
$(p \rightarrow r) \land (q \rightarrow r) \equiv (p \lor q) \rightarrow r$
$(p \rightarrow r) \lor (q \rightarrow r) \equiv (p \land q) \rightarrow r$

Fundamental Proofs

1. **Material Implication:** $p \rightarrow q \equiv \neg p \lor q$

- **Definition:** By definition in classical logic, a conditional is false only when p is true and q is false. This is logically equivalent to saying "either p is false or q is true."
- **Truth Table Verification:**

p	q	$p \rightarrow q$	$\neg p$	$\neg p \lor q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

2. **Contrapositive:** $p \rightarrow q \equiv \neg q \rightarrow \neg p$

- **Proof:** $p \rightarrow q \equiv \neg p \lor q$ (Material Implication) $\equiv q \lor \neg p$ (Commutative Law) $\equiv \neg(\neg q) \lor \neg p$ (Double Negation) $\equiv \neg q \rightarrow \neg p$ (Material Implication)

3. **Disjunction as Conditional:** $p \lor q \equiv \neg p \rightarrow q$

- **Proof:** $\neg p \rightarrow q \equiv \neg(\neg p) \lor q$ (Material Implication) $\equiv p \lor q$ (Double Negation)

4. **Conjunction as Negated Conditional:** $p \land q \equiv \neg(p \rightarrow \neg q)$

- **Proof:** $\neg(p \rightarrow \neg q) \equiv \neg(\neg p \lor \neg q)$ (Material Implication) $\equiv \neg(\neg p) \land \neg(\neg q)$ (De Morgan's Law) $\equiv p \land q$ (Double Negation)

5. **Negation of Conditional:** $\neg(p \rightarrow q) \equiv p \land \neg q$

- **Proof:** $\neg(p \rightarrow q) \equiv \neg(\neg p \lor q)$ (Material Implication) $\equiv \neg(\neg p) \land \neg q$ (De Morgan's Law) $\equiv p \land \neg q$ (Double Negation)

6. $(p \rightarrow q) \land (p \rightarrow r) \equiv p \rightarrow (q \land r)$

- **Proof:** $(\neg p \lor q) \land (\neg p \lor r)$ (Material Implication) $\equiv \neg p \lor (q \land r)$ (Distributive Law: factor out $\neg p$) $\equiv p \rightarrow (q \land r)$ (Material Implication)

7. $(p \rightarrow q) \lor (p \rightarrow r) \equiv p \rightarrow (q \lor r)$

- **Proof:** $(\neg p \lor q) \lor (\neg p \lor r)$ (Material Implication) $\equiv \neg p \lor q \lor \neg p \lor r$ (Associative/Commutative) $\equiv \neg p \lor (q \lor r)$ (Idempotent Law: $\neg p \lor \neg p \equiv \neg p$) $\equiv p \rightarrow (q \lor r)$ (Material Implication)

8. $(p \rightarrow r) \land (q \rightarrow r) \equiv (p \lor q) \rightarrow r$

- **Proof:** $(\neg p \lor r) \land (\neg q \lor r)$ (Material Implication) $\equiv (\neg p \land \neg q) \lor r$ (Distributive Law: factor out r) $\equiv \neg(p \lor q) \lor r$ (De Morgan's Law) $\equiv (p \lor q) \rightarrow r$ (Material Implication)

9. $(p \rightarrow r) \lor (q \rightarrow r) \equiv (p \land q) \rightarrow r$

- **Proof:** $(\neg p \lor r) \lor (\neg q \lor r)$ (Material Implication) $\equiv \neg p \lor r \lor \neg q \lor r$ (Associative/Commutative) $\equiv (\neg p \lor \neg q) \lor r$ (Idempotent Law: $r \lor r \equiv r$) $\equiv \neg(p \land q) \lor r$ (De Morgan's Law) $\equiv (p \land q) \rightarrow r$ (Material Implication)

Logical Equivalences

Biconditionals
$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$
$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

1. Definition: $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

- **Proof:** By definition, a biconditional is true if and only if both components imply each other. $(p \rightarrow q) \wedge (q \rightarrow p) \equiv (\neg p \vee q) \wedge (\neg q \vee p)$ This expression is true only when p and q share the same truth value, which is the definition of $p \leftrightarrow q$.

2. Contrapositive Equivalence: $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$

- **Proof:** $\neg p \leftrightarrow \neg q \equiv (\neg p \rightarrow \neg q) \wedge (\neg q \rightarrow \neg p)$ (Definition) $\equiv (q \rightarrow p) \wedge (p \rightarrow q)$ (Contrapositive Law: $\neg A \rightarrow \neg B \equiv B \rightarrow A$) $\equiv (p \rightarrow q) \wedge (q \rightarrow p)$ (Commutative Law) $\equiv p \leftrightarrow q$ (Definition)

3. Disjunctive Normal Form: $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$

- **Proof:** Start with the definition from #1: $(p \rightarrow q) \wedge (q \rightarrow p) \equiv (\neg p \vee q) \wedge (\neg q \vee p)$ Apply Distributive Law (expand): $\equiv [(\neg p \vee q) \wedge \neg q] \vee [(\neg p \vee q) \wedge p]$ Distribute again: $\equiv [(\neg p \wedge \neg q) \vee (q \wedge \neg q)] \vee [(\neg p \wedge p) \vee (q \wedge p)]$ Apply Complement Law ($A \wedge \neg A \equiv \text{False}$): $\equiv [(\neg p \wedge \neg q) \vee \text{False}] \vee [\text{False} \vee (p \wedge q)]$ Apply Identity Law: $\equiv (\neg p \wedge \neg q) \vee (p \wedge q) \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$ (Commutative Law)

4. Negation of Biconditional: $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

- **Proof:** Using the result from #3: $\neg(p \leftrightarrow q) \equiv \neg[(p \wedge q) \vee (\neg p \wedge \neg q)]$ Apply De Morgan's Law: $\equiv \neg(p \wedge q) \wedge \neg(\neg p \wedge \neg q)$ Apply De Morgan's Law again: $\equiv (\neg p \vee \neg q) \wedge (p \vee q)$ Now expand $p \leftrightarrow \neg q$: $p \leftrightarrow \neg q \equiv (p \rightarrow \neg q) \wedge (\neg q \rightarrow p) \equiv (\neg p \vee \neg q) \wedge (\neg(\neg q) \vee p) \equiv (\neg p \vee \neg q) \wedge (q \vee p) \equiv (\neg p \vee \neg q) \wedge (p \vee q)$ Both expressions result in $(\neg p \vee \neg q) \wedge (p \vee q)$, proving they are equivalent. This represents the "Exclusive OR" (XOR) condition where p and q have different truth values.

Logical Equivalences

Applications

Digital Logic Design: to simplify Boolean expressions, in designing simpler, faster, and more efficient digital circuits.

Software Engineering: to simplify conditional statements and logical expressions in programs, making the code easier to read, understand, and optimize.

Theoretical Computer Science: to analyze algorithms, prove logical correctness, and improve computational efficiency.

Control Systems: to simplify logical conditions, helping control systems work more accurately, efficiently, and reliably.

1. **Which of the following is a tautology?**

   $p \wedge \neg p$ ✅ $p \vee \neg p$ $p \rightarrow \neg p$ $\neg p \rightarrow p$

2. **If the statement "$p \wedge q$" is true, which of the following must also be true?**

   p is true ✅ q is false
   p is false None of the above

3. **If $p \leftrightarrow q$ is true, which of the following must also be true?**

   p and q have the same truth value. ✅
   p is true.
   q is false
   p and q have opposite truth values.

4. **If $p \vee q$ is true and $\neg p$ is true, then q must be true.**

   Valid ✅ Invalid
   Valid only if p is true Can't be determined

1. **The Boolean Expression $(p \wedge \sim q) \vee q \vee (\sim p \wedge q)$ is equivalent to**

✅ $p \wedge q$ $p \vee q$ $p \vee \neg q$ $\neg p \wedge q$

2. **$(p \wedge \neg q) \wedge (\neg p \vee q)$ is**

A ✅ A contradiction B A tautology
C Either A or B D Neither A nor B

3. **If $(p \wedge \sim r) \Rightarrow (q \vee r)$ is false and q and r are both false, then p is**

✅ True False True/False Data insufficient

4. **The negation of the compound proposition $p \vee (\neg p \vee q)$ is**

✅ $(p \wedge \neg q) \wedge \neg p$ $(p \wedge \neg q) \vee \neg p$
$(p \vee \neg q) \vee \neg p$ None of these

5. **The inverse of the proposition $(p \wedge$ ✅ $) \Rightarrow r$ is**

$\neg r \Rightarrow \neg p \vee q$ $\neg p \vee q \Rightarrow \neg r$
$r \Rightarrow p \wedge \neg q$ None of the these

6. **The propositions $(p \Rightarrow \neg p) \wedge (\neg p \Rightarrow p)$ is a**

Tautology and Contradiction
Neither Tautology no ✅ ontradiction
Contradiction
Tautology

1. **The Boolean Expression**
(p∧ ~q)∨q∨(~p∧q) is equivalent to

p ∧ q p ∨ q ✅ p ∨ ¬q ¬p ∧ q

$(P \wedge \neg q) \vee q \vee (\neg P \wedge q)$

$\equiv (P \wedge \neg q) \vee q \vee (q \wedge \neg P)$ – Commutative

$\equiv (P \wedge \neg q) \vee q$ – Absorption

$\equiv q \vee (P \wedge \neg q)$ – Commutative

$\equiv (q \vee P) \wedge (q \vee \neg q)$ – Distributive

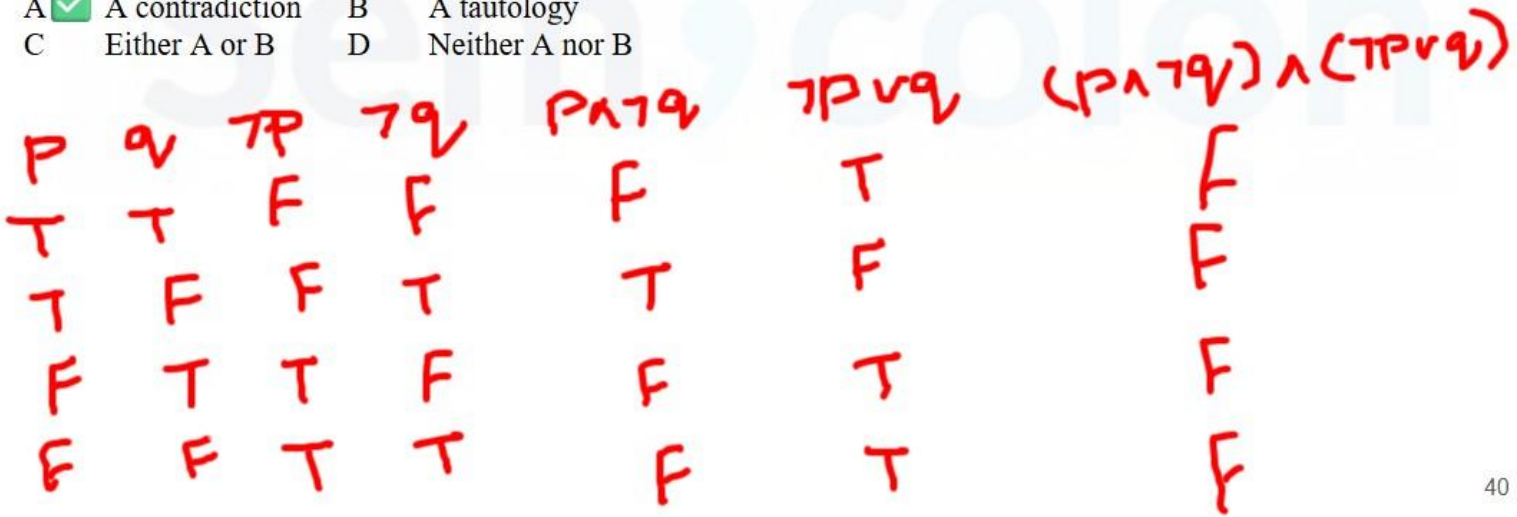
$\equiv (P \vee q) \wedge T$ – Commutative, Negation

$\equiv P \vee q$ – Identity

Semicolon

2. **$(p \wedge \neg q) \wedge (\neg p \vee q)$ is**

A ✅ A contradiction B A tautology
C Either A or B D Neither A nor B



Semicolon

$(p \wedge \neg r) \Rightarrow (q \vee r)$: F

q : F r : F p?

$(p \wedge \neg F) \Rightarrow (F \vee F)$

$(p \wedge T) \Rightarrow F$: F

$p \Rightarrow F$: F

$\therefore$ p is T

or $T \Rightarrow F \equiv F$

3. **If $(p \wedge \sim r) \Rightarrow (q \vee r)$ is false and q and r are both false, then p is**

✅ True False True/False Data insufficient

Semicolon

$\neg[p \vee (\neg p \vee q)]$

$\neg p \wedge \neg(\neg p \vee q)$

$\neg p \wedge \neg(\neg p) \wedge \neg q$

4) $\neg p \wedge p \wedge \neg q$

$(p \wedge \neg q) \wedge \neg p$

5) Inv of $p \to q$: $\neg p \to \neg q$

Inv of $(p \wedge \neg q) \to r$

: $\neg(p \wedge \neg q) \to \neg r$

: $\neg p \vee q \to \neg r$

6) $(p \to \neg p) \wedge (\neg p \to p)$

$(T \to F) \wedge (F \to T)$

$F \wedge T$

F

4. The negation of the compound proposition $p \vee (\neg p \vee q)$ is

- $(p \wedge \neg q) \wedge \neg p$ ✅
- $(p \wedge \neg q) \vee \neg p$
- $(p \vee \neg q) \vee \neg p$
- None of these

5. The inverse of the proposition $(p \wedge \sim q) \Rightarrow r$ is

- $\neg r \Rightarrow \neg p \vee q$
- $\neg p \vee q \Rightarrow \neg r$ ✅
- $r \Rightarrow p \wedge \neg q$
- None of the these

6. The propositions $(p \Rightarrow \neg p) \wedge (\neg p \Rightarrow p)$ is a

- Tautology and Contradiction
- Neither Tautology nor Contradiction
- Contradiction ✅
- Tautology

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1. **Simplify $(p \vee q) \wedge (\neg p \vee q)$ to an equivalent expression using logical laws.**

2. **"It is raining" (r) "The ground is wet" (w) Check if the statement $(r \rightarrow w) \wedge (w \rightarrow r)$ is logically equivalent to $\neg r \vee w$. Justify with truth table**

3. **Which one of the following is not equivalent to p <-> q**

$(\neg p \vee q) \wedge (p \vee \neg q)$ $(\neg p \vee q) \wedge (q \rightarrow p)$ ✅

$(\neg p \wedge q) \vee (p \wedge \neg q)$ $(\neg p \wedge \neg q) \vee$

4. **P: $p \rightarrow (q \vee r)$ and Q: $(p \rightarrow q) \vee (p \rightarrow r)$.**

Are P and Q logically equivalent? Justify using a truth table.

5. **Which sequence of laws is used to prove that**

$p \wedge (p \vee q) \equiv p$?

Commutative Law, Identity Law, Negation Law
Distributive Law, Idempotent Law, Absorption Law ✅
De Morgan's Law, Double Negation Law, Identity Law
Associative Law, Domination Law, Commutative Law

1. **Simplify $(p \vee q) \wedge (\neg p \vee q)$ to an equivalent expression using logical laws.**

$(p \vee q) \wedge (\neg p \vee q)$

$(q \vee p) \wedge (q \vee \neg p)$: Commutative

$q \vee (p \wedge \neg p)$: Reverse distribution

$q \vee F$: Negation Law

q // : Identity Law

Semicolon

2. "It is raining" (r) "The ground is wet" (w)
Check if the statement $(r \rightarrow w) \wedge (w \rightarrow r)$ is logically equivalent to $\neg r \vee w$. Justify with truth table

Not logically equivalent

r	w	$(r \rightarrow w)$	$(w \rightarrow r)$	$(r \rightarrow w) \wedge (w \rightarrow r)$	$\neg r \vee w$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	T
F	F	T	T	T	T

Semicolon

a) $(\neg P \vee q) \wedge (P \vee \neg q)$

$\equiv (P \rightarrow q) \wedge (q \rightarrow P)$

$\equiv P \leftrightarrow q$ //

b) $(\neg P \vee q) \wedge (q \rightarrow P)$

$\equiv (P \rightarrow q) \wedge (q \rightarrow P)$

$\equiv P \leftrightarrow q$ //

c) $(\neg P \wedge q) \vee (P \wedge \neg q)$

$\equiv (\neg P \vee P) \wedge (\neg P \vee \neg q) \wedge (q \vee P) \wedge (q \vee \neg q)$

$\equiv T \wedge (\neg P \vee \neg q) \wedge (P \vee q) \wedge T$

$\equiv (\neg P \vee \neg q) \wedge (P \vee q)$

$\not\equiv P \leftrightarrow q$

d) $(\neg P \wedge \neg q) \vee (P \wedge q)$

$\equiv (\neg P \vee P) \wedge (\neg P \vee q) \wedge (\neg q \vee P) \wedge (\neg q \vee q)$

$\equiv T \wedge (\neg P \vee q) \wedge (\neg q \vee P) \wedge T$

$\equiv (P \rightarrow q) \wedge (q \rightarrow P)$

$\equiv (P \leftrightarrow q)$ //

3. **Which one of the following is not equivalent to p <-> q**

$(\neg p \vee q) \wedge (p \vee \neg q)$	$(\neg p \vee q) \wedge (q \rightarrow p)$
$(\neg p \wedge q) \vee (p \wedge \neg q)$ ✅	$(\neg p \wedge \neg q) \vee (p \wedge q)$

Semicolon

4. P: p→(q∨r) and Q: (p→q)∨(p→r).

Are P and Q logically equivalent? Justify using a truth table.



Semicolon

4. **P: p→(q∨r) and Q: (p→q)∨(p→r).**

   Are P and Q logically equivalent? Justify using a truth table.

$P \wedge (P \vee q)$

$\equiv (P \wedge P) \vee (P \wedge q)$ → Distributive

$\equiv P \vee (P \wedge q)$ → Idempotent

$\equiv P$ → Absorption

5. **Which sequence of laws is used to prove that**

   $p \wedge (p \vee q) \equiv p$?

Commutative Law, Identity Law, Negation Law
Distributive Law, Idempotent Law, Absorption Law ✅
De Morgan's Law, Double Negation Law, Identity Law
Associative Law, Domination Law, Commutative Law

1. **Prove that $(p \rightarrow q) \land (p \rightarrow \neg q) \equiv \neg p$.**

2. **P: "If I study, I will pass." $(s \rightarrow p)$**
Q: "If I do not pass, then I did not study." $(\neg p \rightarrow \neg s)$
R: "If I pass, then I studied." $(p \rightarrow s)$
Which of the following is true?

P and Q are equivalent, ✅ but not R
P and R are equivalent, but not Q
All three are equivalent
None are equivalent

3. **Prove that $p \rightarrow (q \rightarrow r) \equiv (p \land q) \rightarrow r$.**

4. **Show that $(p \lor q) \land (p \lor r) \equiv p \lor (q \land r)$.**

5. **To prove $(p \land q) \rightarrow r \equiv p \rightarrow (q \rightarrow r)$, which combination of laws is applied in the correct order?**
De Morgan's, Distributive, Commutative, Implication
Implication, De Morgan's, Associative, Implication twice ✅
Absorption, Idempotent, Negation, Implication
Double Negation, Commutative, De Morgan's, Identity

1. **Prove that $(p \rightarrow q) \wedge (p \rightarrow \neg q) \equiv \neg p$.**

$LHS = (P \rightarrow q) \wedge (P \rightarrow \neg q)$

$\equiv P \rightarrow (q \wedge \neg q)$

$\equiv P \rightarrow F$

$\equiv \neg P \vee F$

$\equiv \neg P = RHS$

2. **P: "If I study, I will pass." $(s \rightarrow p)$**
Q: "If I do not pass, then I did not study." $(\neg p \rightarrow \neg s)$
R: "If I pass, then I studied." $(p \rightarrow s)$
Which of the following is true?

- P and Q are equivalent, but not R ✅
- P and R are equivalent, but not Q
- All three are equivalent
- None are equivalent

P: $s \rightarrow p$: Implication
Q: $\neg p \rightarrow \neg s$: Contrapositive } Equiv
R: $p \rightarrow s$: Converse

3. **Prove that $p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$.**

$LHS = p \rightarrow (q \rightarrow r) \equiv p \rightarrow (\neg q \vee r)$

$\equiv \neg p \vee (\neg q \vee r) \equiv (\neg p \vee \neg q) \vee r$

$\equiv \neg (p \wedge q) \vee r \equiv (p \wedge q) \rightarrow r$

$= RHS$

4. **Show that $(p \vee q) \wedge (p \vee r) \equiv p \vee (q \wedge r)$.**

$LHS = (p \vee q) \wedge (p \vee r) \equiv (p \wedge p) \vee (p \wedge r) \vee (q \wedge p) \vee (q \wedge r)$

$\equiv p \vee (p \wedge r) \vee (p \wedge q) \vee (q \wedge r)$

$\equiv p \vee (p \wedge q) \vee (q \wedge r) \equiv p \vee (q \wedge r) = RHS$

5. **To prove $(p \wedge q) \rightarrow r \equiv p \rightarrow (q \rightarrow r)$, which combination of laws is applied in the correct order?**

- De Morgan's, Distributive, Commutative, Implication
- Implication, De Morgan's, Associative, Implication twice ✅
- Absorption, Idempotent, Negation, Implication
- Double Negation, Commutative, De Morgan's, Identity

$(p \wedge q) \rightarrow r \equiv \neg (p \wedge q) \vee r$

$\equiv (\neg p \vee \neg q) \vee r \equiv \neg p \vee (\neg q \vee r)$

$\equiv \neg p \vee (q \rightarrow r) \equiv p \rightarrow (q \rightarrow r)$

NET November 2017 Paper 3

Let P, Q, R and S be Propositions. Assume that the equivalences $P \Leftrightarrow (Q \vee \neg Q)$ and $Q \Leftrightarrow R$ hold.

Then the truth value of the formula $(P \wedge Q) \Rightarrow ((P \wedge R) \vee S)$ is always:

True ✅ False Same as truth table of Q Same as truth table of S

GATE 2008

P and Q are two propositions. Which of the following Logical expressions are equivalent?

1. $P \vee \neg Q$
2. $\neg(\neg P \wedge Q)$
3. $(P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q)$
4. $(P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge Q)$

Only 1 and 2 Only 1, 2 and 3 ✅

Only 1, 2 and 4 All of these

NET November 2017 Paper 3

Let P, Q, R and S be Propositions. Assume that the equivalences P⇔(Q∨¬Q) and Q⇔R hold.

Then the truth value of the formula (P∧Q)⇒((P∧R)∨S) is always:

True ✅ False Same as truth table of Q Same as truth table of S

$[P \equiv (Q \vee \neg Q) \equiv T]$
$[Q \equiv R]$

$(P \wedge Q) \rightarrow ((P \wedge R) \vee S) \equiv [T \wedge R] \rightarrow [(T \wedge R) \vee S]$
$\equiv R \rightarrow (R \vee S) \equiv \neg R \vee (R \vee S) \equiv (\neg R \vee R) \vee S$
$\equiv T \vee S \equiv T$ //

GATE 2008

P and Q are two propositions. Which of the following Logical expressions are equivalent?

1. $P \vee \neg Q \equiv Q \rightarrow P$ ✓
2. $\neg(\neg P \wedge Q) \equiv P \vee \neg Q \equiv Q \rightarrow P$ ✓
3. $(P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q) \equiv P \wedge (Q \vee \neg Q) \vee (\neg P \wedge \neg Q) \equiv P \wedge T \vee (\neg P \wedge \neg Q) \equiv (P \vee \neg P) \wedge (P \vee \neg Q) \equiv T \wedge (P \vee \neg Q) \equiv Q \rightarrow P$ ✓
4. $(P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge Q) \equiv P \wedge (Q \vee \neg Q) \vee (\neg P \wedge Q) \equiv P \wedge T \vee (\neg P \wedge Q) \equiv (P \vee \neg P) \wedge (P \vee Q) \equiv T \wedge (P \vee Q) \equiv P \vee Q$ ✗

Only 1 and 2 — Only 1, 2 and 3 ✅
Only 1, 2 and 4 — All of these

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GATE 2014

Which one of the following boolean expression is NOT a tautology?

((a -> b) ^ (b -> c)) -> (a -> c)

(a <-> c) -> (~b -> (a ^ c)) ✅

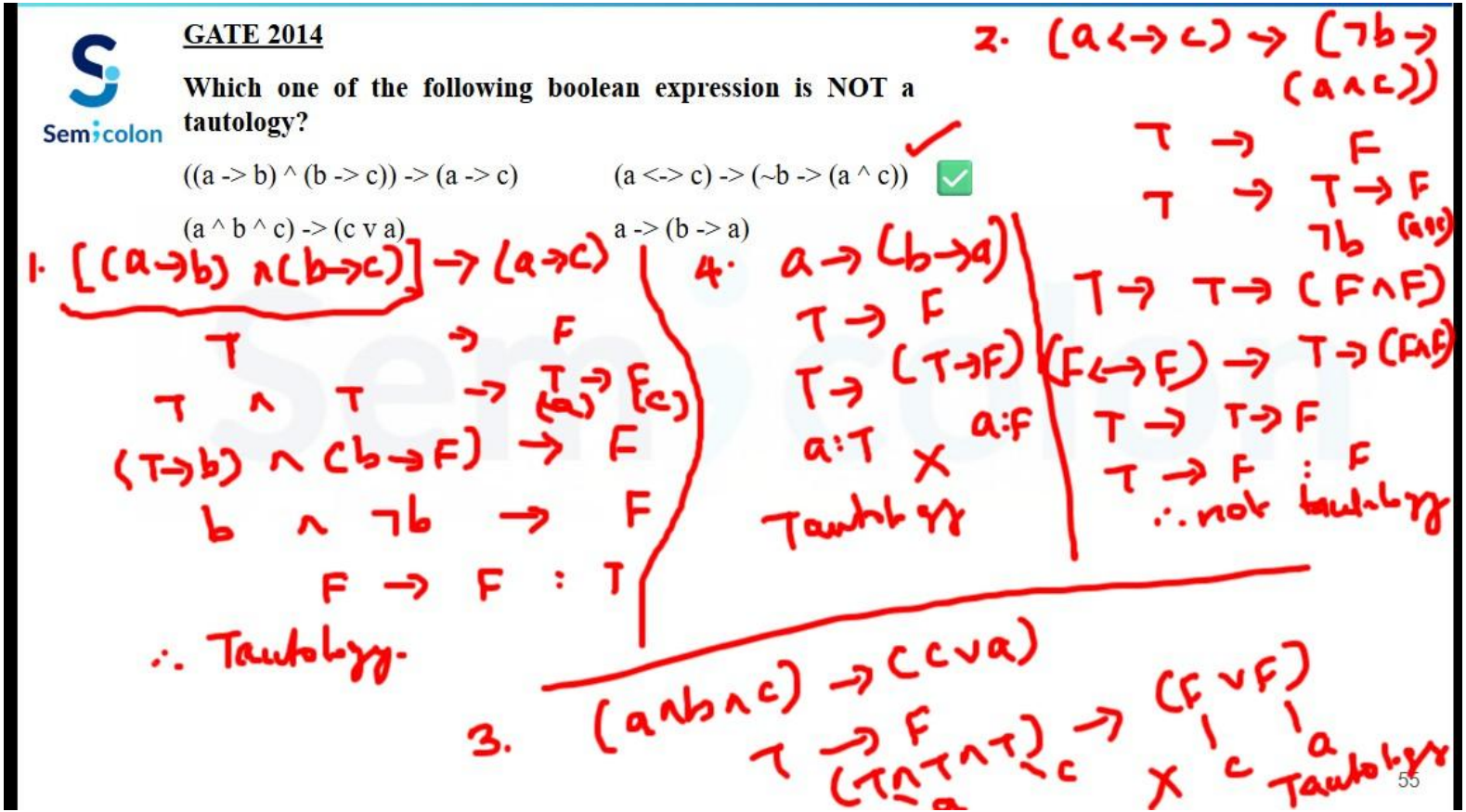
(a ^ b ^ c) -> (c v a)

a -> (b -> a)

GATE 2004

The following propositional statement is
(P -> (Q v R)) -> ((P ^ Q) -> R)

(a) satisfiable but not valid ✅
(b) valid
(c) a contradiction
(d) None of the above


GATE 2014
Which one of the following boolean expression is NOT a tautology?
((a -> b) ^ (b -> c)) -> (a -> c)
(a <-> c) -> (~b -> (a ^ c))
(a ^ b ^ c) -> (c v a)
a -> (b -> a)
1. [(a→b) ∧ (b→c)] → (a→c)
T → F
T ∧ T → T → F
(T→b) ∧ (b→F) → F
b ∧ ¬b → F
F → F : T
∴ Tautology.
4. a → (b→a)
T → F
T → (T→F)
T → F
a:T
a:F
X
Tautology
2. (a<→c) → (¬b → (a∧c))
T → F
T → T → F
¬b
T → T → (F∧F)
(F↔F) → T → (F∧F)
T → T → F
T → F : F
∴ not tautology
3. (a∧b∧c) → (c∨a)
T → F
(T∧T∧T) → (F∨F)
c
a
X
Tautology

GATE 2014

Which one of the following boolean expression is NOT a tautology?

((a -> b) ^ (b -> c)) -> (a -> c)

(a <-> c) -> (~b -> (a ^ c)) ✅

(a ^ b ^ c) -> (c v a)

a -> (b -> a)

check for contradiction:

$[P \to (Q \vee R)] \to [(P \wedge Q) \to R]$

T → F

∴ Invalid

$[T \to (T \vee F)] \to [T \to F]$

$[T \to T] \to [(T \wedge T) \to F]$

T → F : F

GATE 2004

The following propositional statement is
(P -> (Q v R)) -> ((P ^ Q) -> R)

(a) satisfiable but not valid ✅
(b) valid
(c) a contradiction
(d) None of the above

Check for satisfiable: $[P \to (Q \vee R)] \to [(P \wedge Q) \to R]$

F → T

$[T \to F] \to (T \wedge F) \to F$

∴ satisfiable

$[T \to (F \vee F)] \to [F \to F]$

∴ satisfiable but not valid //

F → T : T

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GATE 2005

Define X as (P ∨ Q) → R and Y as (P → R) ∨ (Q → R).
Which one of the following is a tautology?

X ≡ Y ✅ X → Y Y → X ~Y → X

GATE 2009

Which one of the following is equivalent to P ∨ Q ?

(a) ~Q □ ~P
(b) P □ ~Q ✅
(c) ~P □ Q
(d) ~P □ ~Q

P	Q	P □ Q
T	T	T
T	F	T
F	T	F
F	F	T

GATE 2005

Define X as (P ∨ Q) → R and Y as (P → R) ∨ (Q → R). Which one of the following is a tautology?

X ≡ Y X → Y Y → X ~Y → X

a) $(A \wedge B) \not\equiv (A \vee B)$

$\therefore X \neq Y$

b) $(A \wedge B) \rightarrow (A \vee B)$

$T \rightarrow F$

$(F \wedge F) \rightarrow (F \vee F)$

Not possible. ∴ Tautology.

c) $(A \vee B) \rightarrow (A \wedge B)$

$T \rightarrow F$

A: T B: F $T \vee F \rightarrow T \wedge F : F$

$X : (P \vee Q) \rightarrow R$

$\therefore \neg(P \vee Q) \vee R$

$\therefore (\neg P \wedge \neg Q) \vee R$

$\therefore (\neg P \vee R) \wedge (\neg Q \vee R)$

$\therefore (P \rightarrow R) \wedge (Q \rightarrow R)$

$Y : (P \rightarrow R) \vee (Q \rightarrow R)$

X : A ∧ B

Y : A ∨ B

d) $\neg Y \rightarrow X$

$\neg(A \vee B) \rightarrow A \wedge B$

$T \rightarrow F$

A: T B: F $\neg(T \vee F) \rightarrow T \wedge F$

$\neg T \rightarrow F$

$F \rightarrow F : T$

Semicolon

P	Q	¬P	¬Q	P ∨ Q	a) ¬Q → ¬P	b) P → ¬Q	c) ¬P → Q	d) ¬P ↔ ¬Q
T	T	F	F	T	T	T	F	T
T	F	F	T	T	T	T	T	F
F	T	T	F	T	F	T	T	T
F	F	T	T	F	T	F	T	T

GATE 2009

Which one of the following is equivalent to P ∨ Q ?

(a) ~Q □ ~P
(b) P □ ~Q ✅
(c) ~P □ Q
(d) ~P □ ~Q

P	Q	P □ Q
T	T	T
T	F	T
F	T	F
F	F	T

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Logical Implication

The compound propositions p and q are logical implication **if p→q is a tautology.**
Denoted by **p ⇒ q**

GATE 2016

Consider the following expressions:

i. False ii. Q iii. True iv. P ∨ Q v. ¬Q ∨ P

The number of expressions given above that are logically implied by P ∧ (P -> Q) is?

Logical Implication

The compound propositions p and q are logical implication **if p→q is a tautology.**
Denoted by **p ⇒ q**

$P \wedge (P \rightarrow Q) : P \wedge (\neg P \vee Q) : (P \wedge \neg P) \vee (P \wedge Q)$
$: F \vee (P \wedge Q) : (P \wedge Q)$

i) $(P \wedge Q) \rightarrow F$
✗ $T \rightarrow F : F$
$T \wedge T \rightarrow F : F$

ii) $(P \wedge Q) \rightarrow Q$ $\quad Q : F$
$T \rightarrow F$ $\quad P \wedge F : F$
$(P \wedge F) \rightarrow F : T$

iii) $(P \wedge Q) \rightarrow T : T$

iv) $(P \wedge Q) \rightarrow (P \vee Q)$
$T \rightarrow F$
$T \wedge T \rightarrow T \vee T : T$

v) $(P \wedge Q) \rightarrow (\neg Q \vee P)$
✓ $T \rightarrow F$ | $(T \wedge T) \rightarrow (F \vee T)$
$T \rightarrow T \neq F$

GATE 2016

Consider the following expressions:

i. False ii. Q iii. True iv. P ∨ Q v. ¬Q ∨ P

The number of expressions given above that are logically implied by P ∧ (P -> Q) is?

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Rules of Inference in Propositional Logic

Basic Terminology and Examples

Premise:

- A proposition on the basis of which we would able to draw a conclusion.
- An Evidence or assumption.
- Initially we assume something is true and on the basis of that assumption, we draw some conclusion.

Conclusion:

- A proposition that is reached from the given set of premises.
- Result of the assumptions that was made in an argument.

if premise then conclusion

Rules of Inference in Propositional Logic

Basic Terminology and Examples

Argument:

- Sequence of statements that ends with a conclusion.
- Set of one or more premises and a conclusion.

Valid Argument

An argument is said to be valid if and only if it is not possible to make all premises true and a conclusion false.

p: I love cat **q**: I love dog

P1: "If I love cat then I love dog." $p \rightarrow q$

P2: "I love cat." p

C: Therefore, "I love dog." $\therefore q$

Logical Form: $[(p \rightarrow q) \wedge p] \rightarrow q$

Invalid Argument

An argument is said to be invalid if and only if it is possible to make all premises true and a conclusion false.

p: I love cat **q**: I love dog

P1: "If I love cat then I love dog." $p \rightarrow q$

P2: "I love dog." q

C: Therefore, "I love cat." $\therefore p$

$[(p \rightarrow q) \wedge q] \rightarrow p$

Rules of Inference in Propositional Logic

Definition and Types of Inference Rules

Inference: Deriving conclusions from evidences.

Rules of Inference: Templates for constructing valid arguments.

Types of Inference Rules:

1. Modus Ponens
(Rule of detachment)

$$\frac{p \rightarrow q \quad p}{\therefore q}$$

$$[(p \rightarrow q) \wedge p] \rightarrow q$$

2. Modus Tollens

$$\frac{p \rightarrow q \quad \neg q}{\therefore \neg p}$$

$$[(p \rightarrow q) \wedge \neg q] \rightarrow \neg p$$

3. Hypothetical Syllogism
(Transitive)

$$\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$$

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$$

4. Disjunctive Syllogism

$$\frac{p \vee q \quad \neg p}{\therefore q}$$

$$[(p \vee q) \wedge \neg p] \rightarrow q$$

Rules of Inference in Propositional Logic

Definition and Types of Inference Rules

Types of Inference Rules:

5. Addition

$$\frac{p}{\therefore p \vee q}$$

$p \rightarrow (p \vee q)$

6. Simplification

$$\frac{p \wedge q}{\therefore p}$$

or

$$\frac{p \wedge q}{\therefore q}$$

$(p \wedge q) \rightarrow p$ or $(p \wedge q) \rightarrow q$

7. Conjunction

$$\frac{p \quad q}{\therefore p \wedge q}$$

$[(p) \wedge (q)] \rightarrow (p \wedge q)$

8. Resolution

$$\frac{p \vee q \quad \neg p \vee r}{\therefore q \vee r}$$

$[(p \vee q) \wedge (\neg p \vee r)] \rightarrow (q \vee$

Rules of Inference in Propositional Logic

Building Arguments using Rules of Inference

Premises: "Randy works hard",
"If Randy works hard, then he is a dull boy", and
"If Randy is a dull boy, then he will not get the job".

Conclusion: "Randy will not get the job."

Let **H** = Randy works hard,
D = Randy is a dull boy,
J = Randy will get the Job.

Rules of Inference in Propositional Logic

Building Arguments using Rules of Inference

Premises: "Randy works hard",
"If Randy works hard, then he is a dull boy", and
"If Randy is a dull boy, then he will not get the job".

Conclusion: "Randy will not get the job."

Let **H** = Randy works hard,
D = Randy is a dull boy,
J = Randy will get the Job.

$$\frac{H \quad H \rightarrow D \quad D \rightarrow \neg J}{\therefore \neg J}$$

1) $\dfrac{H \quad H \rightarrow D}{\therefore D}$ (Modus Ponens) $\Rightarrow \dfrac{D \quad D \rightarrow \neg J}{\therefore \neg J}$ (Mod Pon)

2) $\dfrac{H \rightarrow \neg D \quad D \rightarrow \neg J}{\therefore H \rightarrow \neg J}$ (Hy syllo) $\Rightarrow \dfrac{H \quad H \rightarrow \neg J}{\therefore \neg J}$ (Mod Pon)

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Rules of Inference in Propositional Logic

Building Arguments using Rules of Inference

Premises: "If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on"
"If the sailing race is held, then the trophy will be awarded", and
"The trophy was not awarded."

Conclusion: "It rained."

Let **R :** It rains
F : It is foggy
S : The sailing race is held
D : The lifesaving demonstration goes on
T : The trophy is awarded

Semicolon

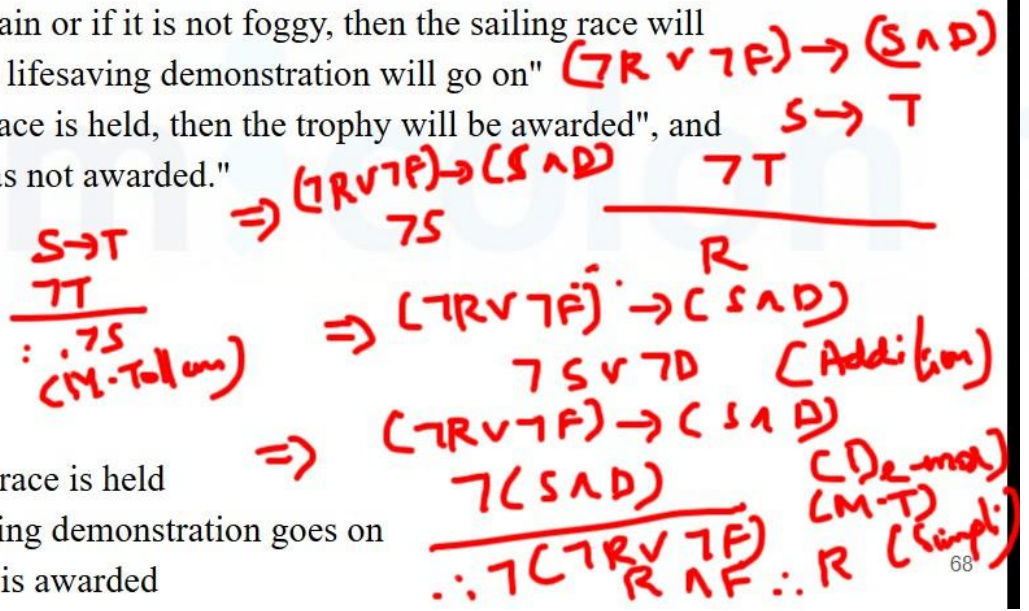
Rules of Inference in Propositional Logic

Building Arguments using Rules of Inference

Premises: "If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on"
"If the sailing race is held, then the trophy will be awarded", and
"The trophy was not awarded."

Conclusion: "It rained."

Let
R : It rains
F : It is foggy
S : The sailing race is held
D : The lifesaving demonstration goes on
T : The trophy is awarded



Rules of Inference in Propositional Logic

Building Arguments using Rules of Inference

Problem: Show that the following argument is valid.

If today is Tuesday, then I have a test in Mathematics or Economics.
If my Economics Professor is sick, then I will not have a test in Economics.
Today is Tuesday and my Economics Professor is sick.
Therefore I have a test in Mathematics.

Solution: Let T = Today is Tuesday, M = I have a test in Mathematics, E = I have a test in Economics and S = My Economics Professor is sick.

Rules of Inference in Propositional Logic

Building Arguments using Rules of Inference

Problem: Show that the following argument is valid.

If today is Tuesday, then I have a test in Mathematics or Economics.
If my Economics Professor is sick, then I will not have a test in Economics.
Today is Tuesday and my Economics Professor is sick.
Therefore I have a test in Mathematics.

$T \rightarrow (M \vee E)$
$S \rightarrow \neg E$
$T \wedge S$
$\therefore M$

Solution: Let T = Today is Tuesday, M = I have a test in Mathematics, E = I have a test in Economics and S = My Economics Professor is sick.

$T \rightarrow (M \vee E)$ T
$S \rightarrow \neg E$ T
$T \wedge S$ T
$\therefore M$ $\therefore ?$ (T)

$T \wedge S : T$
$\Rightarrow T : T \quad S : T$
$S \rightarrow \neg E : T$
$T \rightarrow \neg E : T$
$\Rightarrow \neg E : T \quad E : F$

$T \rightarrow (M \vee E) : T$
$T \rightarrow (M \vee F) : T$
$T \rightarrow M : T$
$M : T$
Valid

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Rules of Inference in Propositional Logic

Building Arguments using Rules of Inference

Consider the following logical inferences.

I1: If it rains then the cricket match will not be played.
The cricket match was played.
Inference: There was no rain.

I2: If it rains then the cricket match will not be played.
It did not rain.
Inference: The cricket match was played.

Which of the following is TRUE?

(a) Both I1 and I2 are correct inferences

(b) I1 is correct but I2 is not a correct inference ✅

(c) I1 is not correct but I2 is correct inference

(d) Both I1 and I2 are not correct inferences

Let R = it rains
C = the cricket match was played.

Answer: (b) I1 is correct but I2 is not a correct inference

Analysis

Given: $R \rightarrow \neg C$

I1 — Modus Tollens ✓

Step	Statement	Rule
1	$R \rightarrow \neg C$	Premise
2	C	Premise (i.e., $\neg(\neg C)$)
3	$\neg R$	Modus tollens

Valid. ✓

I2 — Denying the Antecedent ✗

Step	Statement	Rule
1	$R \rightarrow \neg C$	Premise
2	$\neg R$	Premise
3	C	✗ Invalid

Rules of Inference in Propositional Logic

Building Arguments using Rules of Inference

Consider the following two statements.

S1: If a candidate is known to be corrupt, then he will not be elected.

S2: If a candidate is kind, he will be elected.

Which one of the following statements follows from S1 and S2 as per sound inference rules of logic?

(a) if a person is known to be corrupt, he is kind. ✅

(b) if a person is not known to be corrupt, he is not kind.

(c) if a person is kind, he is not known to be corrupt.

(d) if a person is not kind, he is not known to be corrupt.

[GATE 2015 (set-2): 1 Mark]

Let C = candidate is corrupt,

E = candidate will be elected and

K = candidate is kind.

Step	Statement	Justification
1	$C \rightarrow \neg E$	S1
2	$K \rightarrow E$	S2
3	$E \rightarrow \neg C$	Contrapositive of S1
4	$K \rightarrow \neg C$	**Hypothetical syllogism** on (2) and (3)

Step 4 reads: *If a person is kind, he is not known to be corrupt* → **Option (c)** ✓

Why the others fail

Option	Form	Error
(a) $C \rightarrow K$	No valid chain leads here	—
(b) $\neg C \rightarrow \neg K$	Inverse of $K \rightarrow \neg C$	**Denying the antecedent**
(d) $\neg K \rightarrow \neg C$	Converse of $K \rightarrow \neg C$	**Affirming the consequent**

Limitation of Propositional Logic

The argument which propositional logic can handle

If it snows today, then we will go skiing.
It is snowing today.

∴ We will go skiing.

Let p denotes "it is snowing today."
q denotes "We will go skiing."

Valid argument as it is true by the rule of Modus ponens.

The argument which propositional logic could not handle

Everyone enrolled in the university has lived in a dormitory.
Mia has never lived in a dormitory.

∴ Mia is not enrolled in the university.

No Rules of propositional logic allow us to conclude the truth of r.
Therefore, we need a more powerful type of logic called First Order Logic or Predicate Logic.

First Order Logic or Predicate Logic

Predicates

The statements involving variables which are neither True nor False until or unless the values of the variables are specified.

x is an animal *x is greater than 3* *x is less than 4* $x + y = 7$

All these statements are neither True nor False, and are not propositions.

In Predicate Logic, a statement is divided into two parts: **Subject** & **Predicate**

"x is an animal".

x is a subject (something we are discussing about).

"is an animal" is a predicate (refers to a property that the subject of the statement can have).

The statement "x is greater than 3" can be represented by G(x)

where G() denotes the predicate "is greater than 3" and x denotes the subject or variable.

After assigning the value of a variable x, the statement G(x) becomes a proposition and has a truth value (either T or F)

G(5) = 5 is greater than 3 → True

G(3) = 3 is greater than 3 → False

First Order Logic or Predicate Logic

Finding truth values of Predicates

1. Let P(x) denotes the statement *"x > 3"*. What are the truth values of P(4) and P(2)?

   P(x): x > 3 P(4): 4 > 3 (True) P(2): 2 > 3 (False)

2. Let P(x) denotes the statement *"the word x contains the letter 'a'."* What are the truth values of P(Orange), P(Lemon), P(True) and P(False)?

   P(x): the word x contains the letter 'a'.
   P(Orange): the word Or**a**nge contains the letter 'a'. (True)
   P(Lemon): the word Lemon contains the letter 'a'. (False)
   P(True): the word True contains the letter 'a'. (False)
   P(False): the word F**a**lse contains the letter 'a'. (True)

3. Let Q(x, y) denote "x is the capital of y". What are the truth values of Q(Delhi, India), Q(New York, New York), and Q(Denver, Colorado)?

4. If P(x) denotes x < 4, then what are the truth values of P(0), P(6), and P(10)?

Semicolon

First Order Logic or Predicate Logic

Finding truth values of Predicates

1. Let P(x) denotes the statement *"x > 3"*. What are the truth values of P(4) and P(2)?

   P(x): x > 3 P(4): 4 > 3 (True) P(2): 2 > 3 (False)

2. Let P(x) denotes the statement *"the word x contains the letter 'a'."* What are the truth values of P(Orange), P(Lemon), P(True) and P(False)?

   P(x): the word x contains the letter 'a'.
   P(Orange): the word Orange contains the letter 'a'. (True)
   P(Lemon): the word Lemon contains the letter 'a'. (False)
   P(True): the word True contains the letter 'a'. (False)
   P(False): the word False contains the letter 'a'. (True)

4) P(x) : x < 4
P(0) : 0 < 4 : T
P(6) : 6 < 4 : F
P(10) : 10 < 4 : F

3. Let Q(x, y) denote "x is the capital of y". What are the truth values of Q(Delhi, India), Q(New York, New York), and Q(Denver, Colorado)?

   T F T

4. If P(x) denotes x < 4, then what are the truth values of P(0), P(6), and P(10)?

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First Order Logic or Predicate Logic

Quantifiers

Words that refer to quantities such as "some" or "all".
It tells for how many elements a given predicate is True.

In English, Quantifiers are used to express the quantities without giving an exact number.
Ex: all, some, many, none, few etc. "Can I have some water?" "Jack has many friends here."

Types of Quantifiers	
Universal Quantifier	**Existential Quantifier**
Domain of Discourse: Domain of x is set of all +ve integers A domain specifies the possible values of the variable under consideration. Without domain, there is no quantification.	
Universal Quantification of P(x) is the statement **"P(x) for all values of x in the domain."** Let P(x) be the statement: x + 1 > x P(1): 1 + 1 > 1 → 2 > 1 (True) P(2): 2 + 1 > 2 → 3 > 2 (True) …….. P(x) is true for all +ve integers x. **∀ x P(x): For all x, P(x) is true under the domain of +ve integers.** for all x P(x), for every x P(x), for each x P(x), for any x P(x), all of x P(x) **∀ = "for all"**	Existential Quantification of Q(x) is the statement **"There exists an element x in the domain such that Q(x)."** Let Q(x) be the statement: x < 2 Q(1) : 1 < 2 (True) Q(2) : 2 < 2 (False) ……. Is there some value of x for which Q(x) is True? Yes. Q(x) is true for x=1. Therefore, there is an x for which Q(x) is True. **∃ x Q(x) — There exists some x for which Q(x) is true.** there is an x such that Q(x), there is atleast one x such that Q(x), for some x P(x) **∃ = "There exists"**

First Order Logic or Predicate Logic

Universal Quantifiers

When does $\forall x\, P(x)$ become False?

$\forall x\, P(x)$ becomes False, when P(x) is not always True in the domain under consideration.

1. Let P(x) be the statement "x < 4". What is the truth value of the quantification $\forall x\, P(x)$ where the domain consists of all +ve integers?
   Finding just one **counter example** is enough to make $\forall x\, P(x)$ False.
   P(5): 5 < 4 (False)
   As P(5) is False. Therefore, x = 5 is one such counterexample for the statement $\forall x\, P(x)$.
   This shows that P(x) is not True for every x in the domain of +ve integers.

2. Let Q(x) be the statement "x + 1 > 2x". If the domain consists of all integers, what is the truth value of $\forall x\, Q(x)$?
   Q(1): 1 + 1 > 2(1) → 2 > 2 (False)
   Since Q(1) is False, $\forall x\, Q(x)$ is False.

3. If R(x) is the statement "$x^2 \geq 0$", S(x) is the statement "$x^2 \geq x$", and T(x) is the statement "$x^2 < 0$", determine the truth values of $\forall x\, R(x)$, $\forall x\, S(x)$ and $\forall x\, T(x)$ if the domain of each variable consists of all integers.

First Order Logic or Predicate Logic

Universal Quantifiers

3. If R(x) is the statement "$x^2 \geq 0$", S(x) is the statement "$x^2 \geq x$", and T(x) is the statement "$x^2 < 0$", determine the truth values of $\forall x$ R(x), $\forall x$ S(x) and $\forall x$ T(x) if the domain of each variable consists of all integers.

$R(x) : x^2 \geq 0$

$\forall x\, R(x) : T$ [Square of any int is +ve]

$S(x) : x^2 \geq x$

$\forall x\, S(x) : T$ [Square of any int is always $\geq$ the int]

$T(x) : x^2 < 0$

$\forall x\, T(x) : F$ [Square cannot be -ve]

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First Order Logic or Predicate Logic

Expressing Quantifications in English

Let P(x) be the statement "x spends more than five hours every weekday in class." where the domain for x consists of all students. Express $\forall x\ P(x)$ in English.

- For every x in the domain of all students, x spends more than five hours every weekday in class.
- Every student spends more than five hours every weekday in class.

Let N(x) be the statement "x has visited North Korea" where the domain consists of the students in your school. Express $\forall x\ N(x)$ in English.

- For every x in the domain of the students in your school, x has visited North Korea.
- All students in your school have visited North Korea.

What does the statement $\forall x\ P(x)$ mean if P(x) is "x is perfect" and the domain consists of people in your locality?

- For every person in your locality, that person is perfect.
- All people in your locality are perfect.

First Order Logic or Predicate Logic

Existential Quantifiers

What is the truth value of $\exists x\, P(x)$, where P(x) is the statement "$x^2 > 10$" and the universe of discourse consists of the positive integers not exceeding 4? P(x): $x^2 > 10$, Domain: {1, 2, 3, 4}

x	P(x)	Truth Value
1	$1^2 > 10 \rightarrow 1 > 10$	False
2	$2^2 > 10 \rightarrow 4 > 10$	False
3	$3^2 > 10 \rightarrow 9 > 10$	False
4	$4^2 > 10 \rightarrow 16 > 10$	**True**

$\exists x\, P(x) = P(1) \vee P(2) \vee P(3) \vee P(4)$
Since P(4) is True, $\exists x\, P(x)$ is True.

Let P(x) be the statement "$x = x^2$". If the domain consists of the integers. What is the truth value of $\exists x\, P(x)$?

$P(1) : 1 = 1^2$ (True)
So, the truth value of $\exists x\, P(x)$ is True

Let Q(x) be the statement "$x + 1 > 2x$". If the domain consists of all integers, What is the truth value of $\exists x\, Q(x)$?

$Q(x) : x + 1 > 2x$
$Q(-1) : -1 + 1 > 2(-1) \Rightarrow 0 > -2$ (True)
So, $\exists x\, Q(x)$ is True.

Determine the truth value if the domain consists of all real numbers.

a) $\exists x\, (x^3 = -1)$
Let $x = -1$
$(-1)^3 = -1 \rightarrow$ True
So, $\exists x\, (x^3 = -1)$ is True.

b) $\exists x\, (x^4 < x^2)$
Let $x = 1/2$
$(1/2)^4 < (1/2)^2 \Rightarrow 1/16 < 1/4 \rightarrow$ True
So, $\exists x\, (x^4 < x^2)$ is True.

c) $\exists x\, (x^2 = 2)$
Let $x = \sqrt{2}$
$(\sqrt{2})^2 = 2 \rightarrow$ True
So, $\exists x\, (x^2 = 2)$ is True.

d) $\exists x\, (x^2 = -1)$
For any real number x, $x^2 \geq 0$.
There is no real number whose square is negative.
So, $\exists x\, (x^2 = -1)$ is False.

e) $\exists x\, (x^2 < 0)$
For any real number x, $x^2 \geq 0$.
Square can never be negative.
So, $\exists x\, (x^2 < 0)$ is False.

First Order Logic or Predicate Logic

Quantifiers with Restricted Domain

We can limit the domain of the quantifier by modifying the notation a bit.

$\forall x < 0\ (x^2 > 0)$ Domain: Real Numbers
$\forall x < 0\ (x^2 > 0)$ true
Meaning: The Square of a negative real number is positive.
Standard Form: $\forall x < 0\ (x^2 > 0) \Leftrightarrow \mathbf{\forall x\ (x < 0 \rightarrow x^2 > 0)}$

$\exists y > 0\ (y^2 = 2)$ Domain: Real Numbers
$\exists y > 0\ (y = \sqrt{2})$ true.
Meaning: There exists a positive square root of 2.
Standard Form: $\exists y > 0\ (y^2 = 2) \Leftrightarrow \mathbf{\exists y\ (y > 0 \wedge y^2 = 2)}$

What is the truth value of $\forall x < 0\ (x^2 = 1)$ where the domain consists of all real numbers?

Standard form: $\forall x\ (x < 0 \rightarrow x^2 = 1)$
Counterexample: Let $x = -2$. Then $x < 0$ is True, but $x^2 = 4 \neq 1$.
$\forall x < 0\ (x^2 = 1)$ False.

What is the truth value of the following statements if the domain consists of all integers?

$\exists x \neq 2\ (x^2 = 4)$

Standard form: $\exists x\ (x \neq 2 \wedge x^2 = 4)$

Let $x = -2$ $(-2) \neq 2$ and $(-2)^2 = 4$

$\exists x \neq 2\ (x^2 = 4)$ True.

$\exists x\ (x < 0 \rightarrow x^2 - 2 \geq 1)$

The implication is false only when $x < 0$ is True AND $x^2 - 2 \geq 1$ is False (i.e., $x^2 < 3$).

For integers: $x = -1$ gives $x^2 = 1 < 3$, so the implication fails at $x = -1$.

But we only need one x where it's true. Let $x = 0$:

$x < 0$ is False $\rightarrow$ implication is True

$\exists x\ (x < 0 \rightarrow x^2 - 2 \geq 1)$ is True. ✓

First Order Logic or Predicate Logic

Common Quantifier Meanings

Sentence	Abbreviated Meaning
(∀x, P(x)	All True
∃x, P(x)	Atleast One True
¬[∃x, P(x)]	None True
(∀x, [¬P(x)]	All False
∃x, [¬P(x)	Atleast One False
¬[∃x, [¬P(x)]]	None False
¬[(∀x, [P(x)]]	Not All True
¬[(∀x, [¬P(x)]]	Not All False

Predicates vs Quantifiers

Predicate	Quantifier
A statement containing variables	A symbol indicating the scope of the predicate
Describes a property or relation	Specifies the extent to which the predicate is true
P(x): "x is an even number."	$\forall$: "For all" or $\exists$: "There exists"
No specific symbol	$\forall$ (Universal), $\exists$ (Existential)
Used to form logical statements	Used to quantify logical statements
Alone or with quantifiers	Always used with predicates
P(x), Q(x, y)	$\forall x$, $\exists y$
P(x): "x is an even number."	$\forall$ x $\in$N ,P(x) : "For all x in natural numbers, x is even."
Describes properties of individual elements	Extends predicates over a range of elements

Predicates & Quantifiers

Applications in Computer Science

Program Verification:

Used to prove that a program works correctly for all possible inputs and satisfies its specifications.

Databases & Query Languages:

Predicates are used in conditions (WHERE clause), while quantifiers like EXISTS and ALL help filter and retrieve data logically.

Artificial Intelligence (AI):

Helps represent facts, rules, and relationships in knowledge-based systems for logical reasoning and decision-making.

Algorithms and Data Structures:

Used to define preconditions, postconditions, and constraints to ensure algorithm correctness and efficiency.

Compiler Design and Static Analysis:

Compilers use predicates to detect syntax errors, type mismatches, and rule violations before program execution.

Predicates & Quantifiers

Logical Equivalences

Two logical statements involving predicates and quantifiers are said to be equivalent

if and only if they have the same truth value in all the possible cases.

Logical equivalence is helpful in replacing one expression with an equivalent expression.

Universal quantifier distributes over conjunction: $\forall x\,(P(x) \wedge Q(x)) \equiv \forall x\,P(x) \wedge \forall x\,Q(x)$

Domain: All students of xyz university.

P(x): x has studied discrete mathematics.

Q(x): x has scored more than 60% marks in exam.

	Discrete Math	*>60%*
Mark	✓	✓
Joe	✓	✓

$\forall x\,(P(x) \wedge Q(x))$: *Every student in the xyz university has studied discrete mathematics and has scored more than 60% marks in exam.*

$\forall x\,P(x) \wedge \forall x\,Q(x)$: *Every student in the xyz university has studied discrete mathematics and every student in the xyz university has scored more than 60% marks in exam.*

	Discrete Math	*>60%*
Mark	✓	✗
Joe	✗	✓

$\forall x\,(P(x) \wedge Q(x)) \equiv \forall x\,P(x) \wedge \forall x\,Q(x) \rightarrow$ *True*

$\forall x\,(P(x) \vee Q(x)) \equiv \forall x\,P(x) \vee \forall x\,Q(x) \rightarrow$ *False*

Predicates & Quantifiers

Logical Equivalences

Existential quantifier distributes over disjunction: $\exists x\,(P(x) \vee Q(x)) \equiv \exists x\,P(x) \vee \exists x\,Q(x)$

Domain: All students of xyz university.

P(x): x has studied discrete mathematics.

Q(x): x has scored more than 60% marks in exam.

	Discrete Math	>60%
Mark	✓	✗
Joe	✗	✗

	Discrete Math	>60%
Mark	✗	✗
Joe	✗	✓

$\exists x\,(P(x) \vee Q(x))$: *Some students in the xyz university has studied discrete mathematics or has scored more than 60% marks in exam.*

$\exists x\,P(x) \vee \exists x\,Q(x)$: *Some student in the xyz university has studied discrete mathematics or some student in the xyz university has scored more than 60% marks in exam.*

$\exists x\,(P(x) \vee Q(x)) \equiv \exists x\,P(x) \vee \exists x\,Q(x) \rightarrow True$

$\exists x\,(P(x) \wedge Q(x)) \equiv \exists x\,P(x) \wedge \exists x\,Q(x) \rightarrow False$

	Discrete Math	>60%
Mark	✓	✓
Joe	✗	✗

	Discrete Math	>60%
Mark	✓	✗
Joe	✗	✓

Predicates & Quantifiers

Negating Quantified Expression

"Every student in the xyz university has studied discrete mathematics." $\forall x\, P(x)$

Domain: All students of xyz university. P(x): x has studied discrete mathematics.

Negation:

"It is not the case that every student in the xyz university has studied discrete mathematics." $\neg \forall x P(x)$

"There is some student in the xyz university who has not studied discrete mathematics." $\exists x \neg P(x)$

"There is a student in the xyz university who has studied discrete mathematics." $\exists x\, P(x)$

Domain: All students of xyz university. P(x): x has studied discrete mathematics.

Negation:

"It is not the case that there is a student in the xyz university who has studied discrete mathematics." $\neg \exists x P(x)$

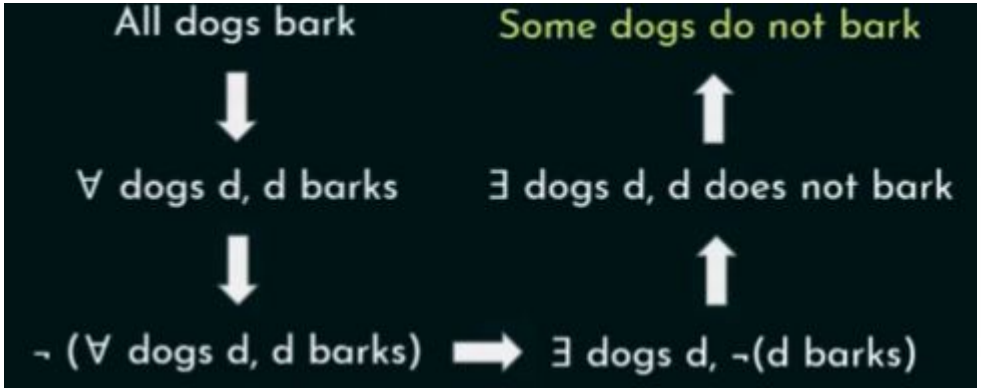
"Every student in the xyz university has not studied discrete mathematics." $\forall x \neg P(x)$

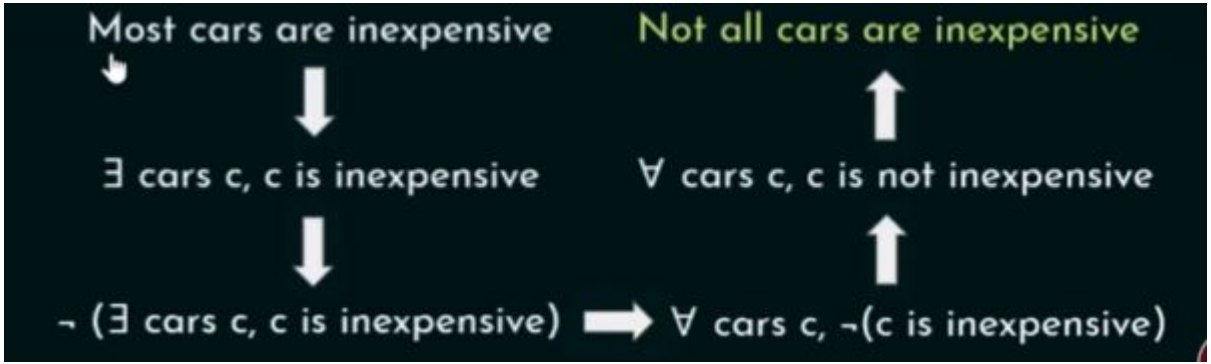
De Morgan's Laws for Quantifiers

$\neg \forall x P(x) \equiv \exists x \neg P(x)$ $\neg \exists x P(x) \equiv \forall x \neg P(x)$

Predicates & Quantifiers

Negating Quantified Expression





All dogs have fleas.

Some dog does not have fleas.

There is a horse that can add.

No horse can add.

No rabbit knows calculus.

Some rabbit knows calculus.

Original	Negation
"All A are B"	"Some A is not B"
"Some A is B"	"No A is B"
"No A is B"	"Some A is B"

Predicates & Quantifiers

Translating English Sentences to Logical Expressions

"Every student of XYZ University has studied discrete mathematics."

Let us suppose that the domain consists of all students of XYZ University.

And, S(x) denotes "x has studied discrete mathematics."

"For every student x in the XYZ University, x has studied discrete mathematics." $\forall x\, S(x)$

Suppose we use a different domain and a set of predicates: Domain: All people on the planet earth.

P(x) denotes "person x is a student in the XYZ University." S(x) denotes "person x has studied discrete mathematics."

"For every person x, if person x is a student in the XYZ University then x has studied discrete mathematics."

$\forall x\, (P(x) \rightarrow S(x))$

Note: We can't write $\forall x\, (P(x) \wedge S(x))$ because this statement says that "All people are students in the XYZ University and has studied discrete mathematics."

Using a two-variable predicate S(x, y) which represents the statement "person x has studied subject y."

$\forall x\, (P(x) \rightarrow S(x,$ discrete mathematics$))$ under the domain of all people on the planet earth.

$\forall x\, (S(x,$ discrete mathematics$))$ — under the domain of all students of XYZ University.

Predicates & Quantifiers

Translating English Sentences to Logical Expressions

"Some student in the XYZ University has studied discrete mathematics."

Let us suppose that the domain consists of all students of XYZ University. P(x) denotes "x has studied discrete mathematics."

"There exists some student x in the XYZ University who has studied discrete mathematics." $\exists x\, P(x)$

Suppose we use a different domain and a set of predicates: Domain: All people on the planet earth.

S(x) denotes "x is a student of XYZ University." P(x) denotes "x has studied discrete mathematics."

"There is some person x, such that x is the student of XYZ University and x has studied discrete mathematics." $\exists x\ (S(x) \wedge P(x))$

1. *"There is a student who has taken more than 21 credit hours in a semester and received all A's"*
2. *Premises:* *"All lions are fierce"* *"Some lions do not drink coffee"*

   Conclusion: *"Some fierce creatures do not drink coffee"* *Domain:* *All creatures*

   Predicates: *P(x): "x is a lion"* *Q(x): "x is fierce"* *R(x): "x drinks coffee"*
3. *Premises:* *"All humming birds are richly colored"* *"No large birds live on honey"*

   "Birds that do not live on honey are dull in color"

   Conclusion: *"Humming birds are small"* *Domain:* *All birds*

Predicates & Quantifiers

Translating English Sentences to Logical Expressions

What is the logical translation of the following statement? "None of my friends are perfect." [GATE CS 2013]

(A) $\exists x(F(x) \wedge \neg P(x))$ (B) $\exists x(\neg F(x) \wedge P(x))$ (C) $\exists x(\neg F(x) \wedge \neg P(x))$ (D) $\neg \exists x(F(x) \wedge P(x))$

Domain: All people in this world. F(x): "x is my friend." P(x): "x is perfect."

"None of my friends are perfect.": $\forall x(F(x) \rightarrow \neg P(x)) \equiv \forall x(\neg F(x) \vee \neg P(x)) \equiv \forall x \neg(F(x) \wedge P(x)) \equiv \neg \exists x(F(x) \wedge P(x))$

Which one of the following is NOT logically equivalent to $\sim \exists x(\forall y(\alpha) \wedge \forall z(\beta))$? [GATE CS 2013]

(A) $\forall x(\exists z(\sim\beta) \rightarrow \forall y(\alpha))$ (B) $\forall x(\forall z(\beta) \rightarrow \exists y(\sim\alpha))$ (C) $\forall x(\forall y(\alpha) \rightarrow \exists z(\sim\beta))$ (D) $\forall x(\exists y(\sim\alpha) \rightarrow \exists z(\sim\beta))$

$\sim \exists x(\forall y(\alpha) \wedge \forall z(\beta)) \equiv \forall x \sim(\forall y(\alpha) \wedge \forall z(\beta)) \equiv \forall x(\sim \forall y(\alpha) \vee \sim \forall z(\beta)) \equiv \forall x(\exists y(\sim\alpha) \vee \exists z(\sim\beta))$

(A) $\forall x(\exists z(\sim\beta) \rightarrow \forall y(\alpha)) \equiv \forall x(\sim \exists z(\sim\beta) \vee \forall y(\alpha)) \equiv \forall x(\forall z(\beta) \vee \forall y(\alpha))$

(B) $\forall x(\forall z(\beta) \rightarrow \exists y(\sim\alpha)) \equiv \forall x(\sim \forall z(\beta) \vee \exists y(\sim\alpha)) \equiv \forall x(\exists z(\sim\beta) \vee \exists y(\sim\alpha))$

Predicates & Quantifiers

Translating English Sentences to Logical Expressions

What is the correct translation of the following statement? "Some real numbers are rational" Domain: All numbers

$\exists$ x(real(x) $\vee$ rational(x)) $\forall$ x(real(x) $\rightarrow$ rational(x)) $\exists$ x(real(x) $\wedge$ rational(x)) $\exists$ x(rational(x) $\rightarrow$ real(x))

(A) There exist some numbers which are either real or rational. **[GATE CS 2012]**

(B) All real numbers are rational.

(C) There exist some numbers which are both real and rational.

(D) There exist some numbers such that if they are rational then they are real.

"Gold and silver ornaments are precious". Domain: All ornaments. [GATE CS 2009]

G(x): x is a gold ornament S(x): x is a silver ornament P(x): x is precious

$\forall$ x(P(x) $\rightarrow$ (G(x) $\wedge$ S(x))) $\forall$ x((G(x) $\wedge$ S(x)) $\rightarrow$ P(x)) $\exists$ x((G(x) $\wedge$ S(x)) $\rightarrow$ P(x)) $\forall$ x((G(x) $\vee$ S(x)) $\rightarrow$ P(x))

(A) For all ornaments x, if x is precious then it is both gold and silver.

(B) For all ornaments x, if x is both gold and silver then it is precious.

(C) For some ornaments x, if x is both gold and silver then it is precious.

Predicates & Quantifiers

Translating English Sentences to Logical Expressions

Let Graph(x) be a predicate which denotes that x is a graph. Let Connected(x) be a predicate which denotes that x is connected. Which of the following first order logic sentences DOES NOT represent the statement:

'Not every graph is connected'?

(A) $\neg \forall x(\text{Graph}(x) \Rightarrow \text{Connected}(x))$ **(B)** $\exists x(\text{Graph}(x) \wedge \neg\text{Connected}(x))$

(C) $\neg \forall x(\neg\text{Graph}(x) \vee \text{Connected}(x))$ **(D)** $\forall x(\text{Graph}(x) \Rightarrow \neg\text{Connected}(x))$ **[GATE CS 2007]**

Solution: "Not every graph is connected" is equivalent to "There is at least one graph which is not connected."

(A) $\neg \forall x(\text{Graph}(x) \Rightarrow \text{Connected}(x))$ represents "It is not the case that for all x, if x is a graph, then it is connected."

OR "It is not the case that every graph is connected."

(B) $\exists x(\text{Graph}(x) \wedge \neg\text{Connected}(x))$ represents "There exists some x such that x is a graph and x is not connected."

OR "There exists some graph which is not connected."

(C) $\neg \forall x(\neg\text{Graph}(x) \vee \text{Connected}(x))$ can be rewritten as $\exists x(\text{Graph}(x) \wedge \neg\text{Connected}(x))$

(D) $\forall x(\text{Graph}(x) \Rightarrow \neg\text{Connected}(x))$ represents "For all x, if x is a graph then it is not connected."

Predicates & Quantifiers

Translating English Sentences to Logical Expressions

The CORRECT formula for the sentence, "not all rainy days are cold" is [GATE CS 2014 (Set 3)]

$\forall$ d(Rainy(d) $\wedge$ $\neg$Cold(d)) $\forall$ d($\neg$Rainy(d) $\rightarrow$ Cold(d)) $\exists$ d($\neg$Rainy(d) $\rightarrow$ Cold(d)) $\exists$ d(Rainy(d) $\wedge$ $\neg$Cold(d))

Solution: "Not all rainy days are cold" can be rewritten as "Some rainy days are not cold"

"Not all rainy days are cold" $\equiv \neg \forall$ d(Rainy(d) $\rightarrow$ Cold(d)) $\equiv \exists$ d$\neg$(Rainy(d) $\rightarrow$ Cold(d))

$\equiv \exists$ d$\neg$($\neg$Rainy(d) $\vee$ Cold(d)) $\equiv \exists$ d(Rainy(d) $\wedge$ $\neg$Cold(d))

English equivalents:

$\forall$ d(Rainy(d) $\wedge$ $\neg$Cold(d)) $\rightarrow$ "Every day is rainy and not cold."

$\forall$ d($\neg$Rainy(d) $\rightarrow$ Cold(d)) $\rightarrow$ "If a day is not rainy, then it is cold." $\rightarrow$ "All non-rainy days are cold."

$\exists$ d($\neg$Rainy(d) $\rightarrow$ Cold(d)) $\rightarrow$ "There exists a day such that if it is not rainy, then it is cold."

$\exists$ d(Rainy(d) $\wedge$ $\neg$Cold(d)) $\rightarrow$ "There exists a day that is rainy and not cold."

"Some boys in the class are taller than all the girls" Note: taller(x,y) is true if x is taller than y. [GATE CS 2004]

(A) ($\exists$ x)(boy(x)$\rightarrow$($\forall$ y)(girl(y)$\wedge$taller(x,y))) (B) ($\exists$ x)(boy(x)$\wedge$($\forall$ y)(girl(y)$\wedge$taller(x,y)))

(C) ($\exists$ x)(boy(x)$\rightarrow$($\forall$ y)(girl(y)$\rightarrow$taller(x,y))) (D) ($\exists$ x)(boy(x)$\wedge$($\forall$ y)(girl(y)$\rightarrow$taller(x,y)))

For universal quantifiers, we use implication and for existential quantifiers, we use conjunction.

Predicates & Quantifiers

Translating English Sentences to Logical Expressions

Tigers and lions attack if they are hungry or threatened. **[GATE CS 2006]**

(A) $\forall x[(tiger(x) \wedge lion(x)) \rightarrow (hungry(x) \vee threatened(x)) \rightarrow attacks(x)]$

(B) $\forall x[(tiger(x) \vee lion(x)) \rightarrow (hungry(x) \vee threatened(x)) \wedge attacks(x)]$

(C) $\forall x[(tiger(x) \vee lion(x)) \rightarrow attacks(x) \rightarrow (hungry(x) \vee threatened(x))]$

(D) $\forall x[(tiger(x) \vee lion(x)) \rightarrow (hungry(x) \vee threatened(x)) \rightarrow attacks(x)]$

(A) $tiger(x) \wedge lion(x)$: an animal is both tiger and lion.

(B) $\forall x[(tiger(x) \vee lion(x)) \rightarrow (hungry(x) \vee threatened(x)) \wedge attacks(x)]$

"For all animals x, if x is either tiger or lion, then they are hungry or threatened, and they attack."

OR "All lions and tigers are hungry or threatened, and attacks."

(C) $\forall x[(tiger(x) \vee lion(x)) \rightarrow attacks(x) \rightarrow (hungry(x) \vee threatened(x))]$

"For all animals x, if x is either tiger or lion, then if they attack, then they are hungry or threatened."

OR "All tigers and lions are hungry or threatened if they attack."

(D) $\forall x[(tiger(x) \vee lion(x)) \rightarrow (hungry(x) \vee threatened(x)) \rightarrow attacks(x)]$

Predicates & Quantifiers

Translating English Sentences to Logical Expressions

If T(x) denotes x is a trigonometric function, P(x) denotes x is a periodic function, then the statement "It is not the case that some trigonometric functions are not periodic" can be logically represented as [ISRO CS 2017]

(A) $\neg\exists x[T(x)\wedge\neg P(x)]$ **(B)** $\neg\exists x[T(x)\vee\neg P(x)]$ **(C)** $\neg\exists x[\neg T(x)\wedge\neg P(x)]$ **(D)** $\neg\exists x[T(x)\wedge P(x)]$

(A) It is not the case that there exist some function which is trigonometric and not periodic.

(B) It is not the case that there exist some function which is trigonometric or not periodic.

(C) It is not the case that there exist some function which is not trigonometric and not periodic.

(D) It is not the case that there exist some function which is trigonometric and periodic.

Q. Let P(x) and Q(x) be arbitrary predicates. Which of the following statements is always TRUE?

(A) $((\forall x(P(x)\vee Q(x))))\Longrightarrow((\forall xP(x))\vee(\forall xQ(x)))$

(B) $(\forall x(P(x)\Longrightarrow Q(x)))\Longrightarrow((\forall xP(x))\Longrightarrow(\forall xQ(x)))$

(C) $(\forall x(P(x))\Longrightarrow\forall x(Q(x)))\Longrightarrow(\forall x(P(x)\Longrightarrow Q(x)))$

(D) $(\forall x(P(x))\Leftrightarrow(\forall x(Q(x))))\Longrightarrow(\forall x(P(x)\Leftrightarrow Q(x)))$ [GATE IT 2005]

Solution: Let us consider all the options one by one, and let's try to make LHS True RHS False so that eventually, the implication becomes false.

Let us consider a domain x with two values x_1 and x_2

(A) $[(\forall x(P(x)\vee Q(x)))]\Longrightarrow[(\forall xP(x))\vee(\forall xQ(x))]$

LHS: True — RHS: False

x	P(x)	Q(x)
x_1	T	F
x_2	F	T

(C) $(\forall x(P(x))\Longrightarrow\forall x(Q(x)))\Longrightarrow(\forall x(P(x)\Longrightarrow Q(x)))$

LHS: True — RHS: False

x	P(x)	Q(x)
x_1	T	F
x_2	F	T

(D) $(\forall x(P(x))\Leftrightarrow(\forall x(Q(x))))\Longrightarrow(\forall x(P(x)\Leftrightarrow Q(x)))$

LHS: True — RHS: False

x	P(x)	Q(x)
x_1	T	F
x_2	F	T

(B) $(\forall x(P(x)\Longrightarrow Q(x)))\Longrightarrow((\forall xP(x))\Longrightarrow(\forall xQ(x)))$

LHS: True — RHS: True

x	P(x)	Q(x)
x_1	T	T
x_2	F	T
x_3	F	F

Nested Quantifiers

Two quantifiers are said to be nested if one is within the scope of the other.

$\forall x \exists y\ Q(x,y)$: $\exists$ is within the scope of $\forall$.

Anything within a scope of the quantifier can be thought of as a propositional function.

$\forall x \exists y\ Q(x,y)$ can be written as $\forall x P(x)$ where $P(x)$: $\exists y\ Q(x,y)$

Different Combinations of Nested Quantifiers

$\forall x \forall y\ Q(x, y)$	$\exists x \exists y\ Q(x, y)$	[Order of quantifiers doesn't matter]
$\forall x \exists y\ Q(x, y)$	$\exists y \forall x\ Q(x, y)$	[Order of quantifiers does matter]

$\forall x \forall y\ Q(x, y) \equiv \forall y \forall x\ Q(x, y)$

$\exists x \exists y\ Q(x, y) \equiv \exists y \exists x\ Q(x, y)$

$\forall x \exists y\ Q(x, y) \neq \exists y \forall x\ Q(x, y)$

Nested Quantifiers

Let x and y be real numbers. P(x,y): x+y=0. Find the truth values of

$\forall x \forall y\ P(x, y)$ $\forall x \exists y\ P(x, y)$ $\exists y \forall x\ P(x, y)$ $\exists x \exists y\ P(x, y)$

For all real numbers x and y, x+y=0----------F

For every real number x, there exist a real number y such that x+y=0----------T

There exist some real number y such that for every real number x, x+y=0----------F

There exist some real numbers x and y such that x+y=0----------T

Let x and y be real numbers. Q(x,y): x.y=0. Find the truth values of

$\forall x \forall y\ Q(x, y)$ $\forall x \exists y\ Q(x, y)$ $\exists y \forall x\ Q(x, y)$ $\exists x \exists y\ Q(x, y)$

For all real numbers x and y, x.y=0----------F

For every real number x, there exist a real number y such that x.y=0----------T

There exist some real number y such that for every real number x, x.y=0----------T

There exist some real numbers x and y such that x.y=0----------T

Let Q(x,y,z): x+y=z. Domain: Real numbers. Find the truth values of $\forall x \forall y \exists z\ Q(x, y, z)$ $\exists z \forall x \forall y\ Q(x, y, z)$

For all real numbers x and y, there is a real number z such that x+y=z----------T

Nested Quantifiers

Translating English Sentences

Steps:

1. Read and understand the given english statement.
2. Find the domain of discourse according to the statement.
3. Find the associated quantifiers.
4. Rewrite the statement so that the domain of discourse and quantifiers are properly visible.
5. Finally, introduce the variables in the statement.

The sum of two positive integers is always positive.

Domain: All integers.

Quantifier: For all ($\forall$)

For every two integers, if the two integers are positive then their sum is also positive.

Variables: Two integers x and y.

For every two integers x and y, if x and y are positive then x=y is also positive.

$\forall x \forall y (((x>0) \wedge (y>0)) \rightarrow (x+y>0))$

A negative real number does not have a square root that is a real number.

Domain: All real numbers.

Quantifier: $\forall$ [All negative real numbers], $\exists$ [A square root]

For all real numbers, if a real number is negative then its square root is not real.

Variables: Two real numbers x and y.

For all real numbers x, if x is negative then it is not the case that there exist some real number y such that $x = y^2$.

$\forall x ((x<0) \rightarrow \neg \exists y (x = y^2))$

Nested Quantifiers *Translating English Sentences*

Every student of XYZ University has a computer or has friend who has a computer.

Domain: Students of XYZ University. Predicates: C(x): x has a computer. F(x,y): x is a friend of y

Quantifier: For all (∀) students, there exist (∃) a friend Variables: Students x and y.

For all students x at XYZ University, x has a computer or there exist some y such that y has a computer and x and y are friends..

$$\forall x\,(C(x) \vee \exists y\,(C(y) \wedge F(x, y)))$$

"Every teacher is liked by some student." **[GATE CS 2005]**

(A) ∀ (x) [teacher(x) → ∃ (y) [student(y) → likes(y,x)]]

(B) ∀ (x) [teacher(x) → ∃ (y) [student(y) ∧ likes(y, x)]]

(C) ∃ (y) ∀ (x) [teacher(x) → [student(y) ∧ likes(y, x)]]

(D) ∀ (x) [teacher(x) ^ ∃ (y) [student(y) → likes(y, x)]]

(A) For all x, if x is a teacher then there exist some y such that if y is a student then y likes x.

(B) For all x, if x is a teacher then there exist some y such that y is a student and y likes x.

$$\forall (x)\ \exists (y)\ [\text{teacher}\ (x) \rightarrow [\text{student}\ (y) \wedge \text{likes}\ (y, x)]]$$

(C) There exist some y such that for every x, if x is a teacher then y is a student and y likes x.

Some student likes all teachers.

Nested Quantifiers

Translating English Sentences

Everyone has exactly one best friend

Domain consisting of all people

Let us suppose that B(x,y) denotes "y is the best friend of x."

"Everyone has exactly one best friend" ——------> $\forall x$(x has exactly one best friend)

$\forall x \exists y(B(x, y))$ —----------> "For every x, there exist some y such that y is the best friend of x.

There could be more than one best friend of x, according to this expression.

Suppose that y is the best friend and there is no person in the world other than y who is the best friend of x.

Let's denote these people by z.

$\forall z ((z \neq y) \rightarrow \neg B(x, z))$ —--------> All people in this world other than y are not best friends of x.

Complete expression —----------> $\forall x \exists y (B(x, y) \wedge \forall z ((z \neq y) \rightarrow \neg B(x, z)))$ —-------> ①

Note: We can also use $\exists !$ which is the **uniqueness quantifier** to represent exactly one.

$\therefore$ we can replace expression 1 by $\forall x \exists !y (B(x, y))$ —----------> ②

Negating Nested Quantifiers

Negation of $\exists x\, P(x)$: $\neg \exists x\, P(x) \equiv \forall x\, \neg P(x)$ Negation of $\forall x\, P(x)$: $\neg \forall x\, P(x) \equiv \exists x\, \neg P(x)$

Negating nested quantifier is as simple as negating single quantifier.

Successively apply the rules of negating single quantifiers to nested quantifiers.

Move the negation inside quantifiers by applying De Morgan's laws for negating single quantifiers.

According to De Morgan's law, $\forall$ will get converted to $\exists$ and $\exists$ will get converted to $\forall$.

Express the negation of the statement $\forall x \exists y\, (xy = 1)$ so that no negation precedes a quantifier.

$\neg \forall x \exists y\, (xy = 1) \equiv \exists x \neg \exists y\, (xy = 1) \equiv \exists x \forall y\, \neg(xy = 1) \equiv \exists x \forall y\, (xy \neq 1)$

Express the statement "Someone has visited every country in the world except Libya" using quantifiers. Then form the negation of the statement so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the words "It is not the case that.")

Let V(x, y) denotes x has visited country y. Domain of x: "all people" Domain of y: "all countries"

"Someone has visited every country except Libya": $\exists x \forall y\, (V(x, y) \leftrightarrow y \neq \text{Libya})$

Negation:: $\forall x \exists y\, (V(x, y) \leftrightarrow y = \text{Libya})$ since $\neg(p \leftrightarrow q) = (p \leftrightarrow \neg q)$

Negating Nested Quantifiers

Express the statement "No one has climbed every mountain in the Himalayas" using quantifiers. Then, form the negation of the statement so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not use the words "It is not the case that.")

Let C(x, y) denotes "x has climbed mountain y in the Himalayas."

"No one has climbed every mountain in the Himalayas": $\neg \exists x \forall y\, C(x, y)$

Negation:: $\neg(\neg \exists x \forall y\, C(x, y)) \equiv \exists x \forall y\, C(x, y)$

There exist some person who has climbed every mountain in the Himalayas.

Express the negation of the proposition "There is a student in this class who has taken every mathematics course offered at this school" using quantifiers, and in English.

T(x, y) = "x has taken mathematics course y"

Original statement: $\exists x \forall y T(x,y)$

Negation: $\neg \exists x \forall y T(x,y) \equiv \forall x \exists y \neg T(x,y)$

"Every student in this class has skipped at least one mathematics course offered at this school."

The Resolution Principle

Used to prove whether an argument is correct or not.

Preliminaries:

- *Literal*: A variable or negation of a variable.
- *Clause*: Disjunction of literals.
- *Resolvant*: For any two clauses C_1 and C_2, if C_1 has a literal L_1 and C_2 has L_2 which is the complement of L_1, then delete L_1 and L_2 from C_1 and C_2 and construct the disjunction of remaining literals. The result so obtained is called the resolvent C of C_1 and C_2.

  Let $C_1 = P \vee Q \vee R$ and $C_2 = \neg P \vee S \vee T$, then the resolvant of C_1 and $C_2 = Q \vee R \vee S \vee T$

Modus Ponens Rule: $[P \wedge (P \rightarrow Q)] \rightarrow Q$

C_1: P C_2: $\neg P \vee Q$ Resolvant C: Q (conclusion)

Modus Tollens Rule: $[\neg Q \wedge (P \rightarrow Q)] \rightarrow \neg P$

C_1: $\neg Q$ C_2: $\neg P \vee Q$ Resolvant C: $\neg P$ (conclusion)

The Resolution Principle

Theorem: Given two clauses C_1 and C_2, a resolvant C of C_1 and C_2 is a logical consequence of C_1 and C_2.

Proof:

Consider two clauses C_1 and C_2. C_1: $L \vee C_1'$ C_2: $\neg L \vee C_2'$ Resolvant C: $C_1' \vee C_2'$

To prove C is the logical consequence of C_1 and C_2.

If C is logical consequence, then if C_1 and C_2 are true then C has to be true.

Assume that C_1 and C_2 are true. For L, there are two possibilities: L is either true or false.

Case 1: L is true. $C_1 : L \vee C_1'$ (True) $C_2 : \neg L \vee C_2'$ (True) C_2' has to be true.

This means $C = C_1' \vee C_2'$ is true because of C_2'

Case 2: L is false. $C_1 : L \vee C_1'$ (True) $C_2 : \neg L \vee C_2'$ (True) C_1' has to be true.

This means $C = C_1' \vee C_2'$ is true because of C_1'.

C is the logical consequence of C_1 and C_2.

Therefore, a resolvent C of C_1 and C_2 is a logical consequence of C_1 and C_2.

The Resolution Principle

How to prove whether an argument is valid

Consider an argument with premises P1, P2, ... , Pr and conclusion C.

To prove whether an argument is valid, put all premises in clause form and add ~C to it.

From the sequence, if □ (empty clause) can be derived, then the argument is valid.

Modus ponens rule: $[P \wedge (P \rightarrow Q)] \rightarrow Q$

In the clause form: C_1: P

C_2: $\neg P \vee Q$

Take C_3 as negation of Q: C3 : ~Q

C_4 : Q (from C_1 and C_2 by resolution)

C_5 : □ (from C_3 and C_4 by resolution)

As an Empty clause is obtained. Therefore, the argument is valid.

The Resolution Principle

Use resolution to show that the hypothesis "Allen is a bad boy or Hillary is a good girl" and "Allen is a good boy or David is happy" imply the conclusion "Hillary is a good girl or David is happy."

Let b denotes "Allen is a good boy" g denotes "Hillary is a good girl"

h denotes "David is happy"

C_1: $\neg b \vee g$ C_2: $b \vee h$ C_3: $\neg(g \vee h) = \neg g \wedge \neg h$

C_4: $\neg g$ C_5: $\neg h$ C_6: $g \vee h$ C_7: h

C_8: $\square$

Determine whether the below argument is valid.

If Superman were able to and willing to prevent evil, he would do so.

If Superman were unable to prevent evil, he would be impotent;

If he were unwilling to prevent evil, he would be malevolent.

Superman does not prevent evil.

If Superman exists, he is neither impotent nor malevolent.

p = "Superman is able to prevent evil"

q = "Superman is willing to prevent evil"

r = "Superman prevents evil"

s = "Superman is impotent"

t = "Superman is malevolent"

u = "Superman exists"

C_1: $(p \wedge q) \rightarrow r = \neg p \vee \neg q \vee r$

C_2: $\neg p \rightarrow s = p \vee s$ C_7: $\neg q \vee r \vee s$

C_3: $\neg q \rightarrow t = q \vee t$ C_8: $r \vee s \vee t$

C_4: $\neg r = \neg r$ C_9: $s \vee t$

C_5: $u \rightarrow \neg(s \wedge t)$

$= \neg u \vee \neg s \vee \neg t$ C_{10}: $\neg u$

Fallacies

There are some common arguments which are invalid. Though at first glance, they seem right. Such a type of argument is known as fallacy.

Fallacy of Affirming the Conclusion:

Example: $[(p \rightarrow q) \wedge q] \rightarrow p$

When p is false and q is true, $p \rightarrow q$ is true but conclusion is false.

Therefore, p is not a logical consequence of $p \rightarrow q$ and q.

Example: If today is Monday, then John will go to work.

John will go to work.

Therefore, today is Monday.

Let p denotes "today is Monday"

and q denotes "John will go to work."

Fallacy of Denying Hypothesis:

Example: $[(p \rightarrow q) \wedge \neg p] \rightarrow \neg q$

When p is false and q is true, $p \rightarrow q$ and $\neg p$ are true but $\neg q$ is false.

Therefore, the argument is invalid.

Example:

If Socrates is human, then Socrates is mortal.

Socrates is not human

$\therefore$ Socrates is not mortal

Let p denotes "Socrates is human."

and q denotes "Socrates is mortal."

Rules of Inference for Quantified Statements

Universal Instantiation: This rule is used to conclude that P(c) is true when $\forall x\, P(x)$ is true.

$$\frac{\forall x\, P(x)}{\therefore\ P(c)}$$

Determine whether the argument "All students in this class understand logic. David is a student in this class. Therefore, David understands logic." is correct or not.

P(x) denotes "x is a student in this class"

Q(x) denotes "x understands logic"

Premises: $\forall x\,(P(x) \rightarrow Q(x))$ (1)

P(David) (2)

By Universal Instantiation from (1)

P(David) → Q(David) (3)

By Modus Ponens from (2) and (3) Q(David)

Universal Generalization: This rule states that $\forall x\, P(x)$ is true, given the premise, P(c) is true for an arbitrary c.

$$\frac{\text{P(c) for an arbitrary c}}{\therefore\ \forall x\, P(x)}$$

If P(c) holds for any arbitrary element c, then we can conclude $\forall x\, P(x)$.

Justify that if $\forall x(P(x) \rightarrow Q(x))$ and $\forall x(Q(x) \rightarrow R(x))$ are true, then $\forall x(P(x) \rightarrow R(x))$ is true, where the domain of all quantifiers are same.

Let c be some arbitrary element.

Premises: $\forall x(P(x) \rightarrow Q(x))$ (1) $\forall x(Q(x) \rightarrow R(x))$ (2)

By Universal Instantiation from (1) $P(c) \rightarrow Q(c)$ (3)

By Universal Instantiation from (2) $Q(c) \rightarrow R(c)$ (4)

Hypothetical Syllogism from (3) and (4) $P(c) \rightarrow R(c)$ (5)

Rules of Inference for Quantified Statements

Existential Instantiation: This rule allows us to conclude that there is some element c for which P(c) is true when $\exists x P(x)$ is true.

$\exists x P(x)$ *[c cannot be arbitrary.]*

$\therefore$ **P(c) for some c**

Existential Generalization: This rule states that $\exists x P(x)$ is true when for a particular element c, P(c) is true.

P(c) for some element c

$\therefore \exists x P(x)$

Show that if $\exists x(P(x) \wedge Q(x))$ is true then $\exists x P(x) \wedge \exists x Q(x)$ is also true.

1. $\exists x(P(x) \wedge Q(x))$ Premise
2. $P(c) \wedge Q(c)$ Existential Instantiation from (1)
3. $P(c)$ Simplification from (2)
4. $\exists x P(x)$ Existential Generalization from (3)
5. $Q(c)$ Simplification from (2)
6. $\exists x Q(x)$ Existential Generalization from (5)

Universal Modus Ponens Rule: Combination of two rules: Universal Instantiation & Modus Ponens

If $\forall x(P(x) \rightarrow Q(x))$ is true and P(c) is true for some element c in the domain of universal quantifier then Q(c) must also be true.

All men are mortal Socrates is a man $\therefore$ Socrates is mortal

Let P(x) denotes "x is a man"and Q(x) denotes "x is mortal".

Premises: 1. $\forall x(P(x) \rightarrow Q(x))$ 2. P(Socrates)

Universal Instantiation from (1): $P(Socrates) \rightarrow Q(Socrates)$(3)

Modus ponens from (2) & (3). Q(Socrates)

Universal Modus Tollens Rule: Combines universal instantiation and modus Tollens.

If $\forall x(P(x) \rightarrow Q(x))$ is true and $\neg Q(c)$ is true, where c is the particular element in the domain the $\neg P(c)$ must also be true.

Rules of Inference for Quantified Statements

All parrots like fruit

Charlie does not like fruit

∴ Charlie is not a parrot.

P(x): "x is a parrot" Q(x): "x likes fruit"

Premises:

1. $\forall x(P(x) \rightarrow Q(x))$
2. $\neg Q(Charlie)$
3. $P(Charlie) \rightarrow Q(Charlie)$ Universal Instantiation
4. $\neg P(Charlie)$ Modus Tollens

For each of the following arguments, determine whether the argument is correct or not.

Every computer science major takes discrete mathematics course. Ria is taking discrete mathematics. Therefore, Ria is a computer science major.

Let P(x) = "x is a computer science major",

Q(x) = "x takes discrete mathematics"

Premises: $\forall x(P(x) \rightarrow Q(x))$ (1) Q(Ria) (2)

Universal Instantiation from (1): $P(Ria) \rightarrow Q(Ria)$ (3)

P(Ria) ← Cannot be derived from (2) and (3)

Ria could be taking discrete math for another reason. So by Fallacy of affirming the consequent, the argument is invalid.

Everyone who eats granola everyday is healthy. Linda is not healthy. Therefore, Linda does not eat granola every day.

Let P(x) = "x eats granola everyday" and Q(x) = "x is healthy"

Premises: $\forall x(P(x) \rightarrow Q(x))$ (1) $\neg Q(Linda)$ (2)

Universal Instantiation from (1): $P(Linda) \rightarrow Q(Linda)$ (3)

Modus Tollens from (2) and (3): $\neg P(Linda)$

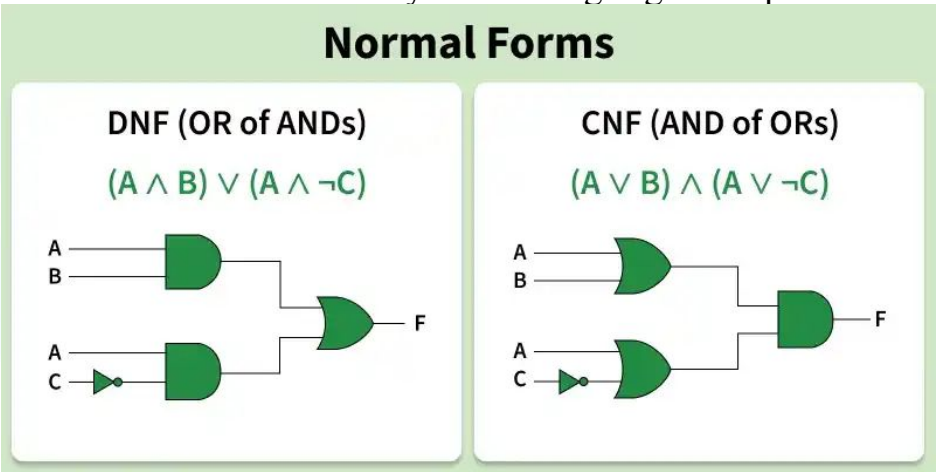
Normal Forms & Principal Forms

In propositional logic, complex logical statements are often converted into standard forms to simplify their analysis and manipulation. Normal forms and principal forms are the standardized ways of writing logical expressions so they are easier to analyze, compare, or implement.

Minterm: Conjunction (AND) of all variables in the expression, where each variable appears exactly once, either in its original form or negated ($A \wedge B \wedge \neg C$).

Maxterm: Disjunction (OR) of all variables in the expression, where each variable appears exactly once, either in its original form or negated ($\neg A \vee B \vee C$).

Normal Forms: Structured representations of logical expressions where the formula is broken down into a combination of literals connected by logical operators such as AND, OR, and NOT.



Disjunctive Normal Form: A logical expression consisting of one or more conjunctions (AND) of literals joined together by disjunctions (OR).

Conjunctive Normal Form: logical expression consisting of one or more disjunctions (OR) of literals joined together by conjunctions (AND).

Normal Forms

If any elementary product in DNF is a tautology, then the given DNF is also a tautology.

$(P \wedge \sim Q) \vee (Q \wedge R) \vee (\sim P \wedge Q \wedge \sim R)$

If every elementary sum in CNF is a tautology, then the given CNF is also a tautology.

$(P \sim \vee Q) \wedge (Q \vee R) \wedge (\sim P \vee Q \vee \sim R)$

The DNF & CNF of any formula are not unique.

Conversion to CNF

Eliminate Bi-conditional and Implication: Convert any bi-conditional and implication into their logical equivalents.

Move Negations Inward: Apply De Morgan's laws to push negations inside and eliminate double negations.

Distribute OR over AND: Apply distributive laws to achieve a conjunction of disjunctions.

Convert $(A \rightarrow B) \wedge \neg C$ to CNF:

$(A \rightarrow B) \wedge \neg C \Rightarrow (\neg A \vee B) \wedge \neg C$

Conversion to DNF

Eliminate Bi-conditional and Implication: Convert any bi-conditional and implication into their logical equivalents.

Move Negations Inward: Apply De Morgan's laws to push negations inside and eliminate double negations.

Distribute AND over OR: Apply distributive laws to achieve a disjunction of conjunctions.

Convert $(A \vee B) \wedge (C \vee D)$ to DNF:

$(A \vee B) \wedge (C \vee D) = [(A \vee B) \wedge C] \vee [(A \vee B) \wedge D]$

$= [(A \wedge C) \vee (B \wedge C)] \vee [(A \wedge D) \vee (B \wedge D)]$

Principal Normal Forms

A Principal Form (PNF) is a canonical representation of a propositional formula in which every term contains all the variables of the expression exactly once, either in its original form or negated. Principal normal forms are constructed using minterms or maxterms obtained from the truth table of the logical expression.

Principal Disjunctive Normal Form (PDNF):

Sum-of-Products Canonical Form

A logical expression consisting of the *disjunction (OR) of minterms* only.

General Structure: $m_1 \vee m_2 \vee \cdots \vee m_n$, where each m_i is a minterm.

Eg: $(P \wedge \sim Q \wedge \sim R) \vee (P \wedge \sim Q \wedge R) \vee (\sim P \wedge \sim Q \wedge \sim R)$

Conversion to PDNF

Construct the Truth Table: Write the truth table for the given logical expression.

Identify the True Rows: Select all rows where the expression evaluates to True.

Write the Corresponding Minterms: For each True row, write a minterm by:

Using the variable itself if its value is True.

Using its negation if its value is False.

Combine the Minterms: Join all the minterms using the OR operator.

*Convert the expression **A** → **B** into PDNF.*

p	q	p → q	Minterms
T	T	T	P ∧ Q
T	F	F	P ∧ ~ Q
F	T	T	~ P ∧ Q
F	F	T	~ P ∧ ~ Q

(P ∧ Q) ∨ (~ P ∧ Q) ∨ (~ P ∧ ~ Q)

Principal Forms

Dual of a minterm is called a maxterm.

Principal Conjunctive Normal Form (PCNF):

Product-of-Sums Canonical Form

A logical expression consisting of the *conjunction (AND) of maxterms* only.

General Structure: $M_1 \wedge M_2 \wedge \cdots \wedge M_n$, where each M_i is a maxterm.

Eg: $(P \vee \sim Q \vee \sim R) \wedge (P \vee \sim Q \vee R) \wedge (\sim P \vee \sim Q \vee \sim R)$

Conversion to PCNF

Construct the Truth Table: Write the truth table for the given logical expression.

Identify the False Rows: Select all rows where the expression evaluates to False.

Write the Corresponding Maxterms: For each False row, write a maxterm by:

Using the variable itself if its value is False.

Using its negation if its value is True.

Combine the Maxterms: Join all the maxterms using the AND operator.

*Convert the expression **A** → **B** into PCNF.*

p	q	p → q	Maxterms
T	T	T	$\sim P \vee \sim Q$
T	F	F	$\sim P \vee Q$
F	T	T	$P \vee \sim Q$
F	F	T	$P \vee Q$

$\sim P \vee Q$

Normal Forms

Properties

Logical Equivalence: Ensures that the logical meaning is preserved

Minimality: Use minimal number of literals and clauses.

Canonical Representation: Unique representation, ensures consistency.

Simplification: Simplify complex logical expressions, easier to manipulate and analyze.

Find the PCNF of $\neg A \vee (B \wedge \neg C)$:

$(\neg A \vee \neg B \vee \neg C) \wedge (\neg A \vee B \vee \neg C) \wedge$

A	T	T	T	T	F	F	F	F
B	T	T	F	F	T	T	F	F
C	T	F	T	F	T	F	T	F
$\neg A \vee (B \wedge \neg C)$	F	T	F	F	T	T	T	T
Maxterm	$\neg A \vee \neg B \vee \neg$		$\neg A \vee B \vee$	$\neg A \vee B \vee$				

Convert $(A \vee B) \wedge (\neg A \vee C)$ *to DNF*

Distribute the AND over OR:

$(A \vee B) \wedge (\neg A \vee C)$

$= (A \wedge \neg A) \vee (A \wedge C) \vee (B \wedge \neg A) \vee (B \wedge C)$

Simplify:

$(A \wedge C) \vee (B \wedge \neg A) \vee (B \wedge C)$

Convert $(A \wedge B) \vee (\neg A \wedge \neg B)$ *to CNF*

Distribute the OR over AND:

$(A \wedge B) \vee (\neg A \wedge \neg B)$

$= (A \vee \neg A) \wedge (A \vee \neg B) \wedge (B \vee \neg A) \wedge (B \vee \neg B)$

Simplify:

$(A \vee \neg B) \wedge (B \vee \neg A)$

Find the PDNF for $A \wedge (\neg B \vee C)$

Step 1: Expand

$A \wedge (\neg B \vee C) = (A \wedge \neg B) \vee (A \wedge C)$

Step 2: Add missing variables

$A \wedge \neg B = (A \wedge \neg B \wedge C) \vee (A \wedge \neg B \wedge \neg C)$

Propositional Logic vs Predicate Logic

Propositional Logic	Predicate Logic
Deals with declarative statements that have a definite truth value	Deals with statements involving variables, predicates, and quantifiers
Basic form of formal Boolean logic	Extends propositional logic and has greater expressive power
Fixed truth value (true or false)	The truth value depends on values assigned to variables
Uses logical connectives	Uses logical connectives along with quantifiers
Does not use variables or quantifiers	Uses variables, functions, and quantifiers
Less expressive but simpler to evaluate	More expressive and suitable for complex reasoning

Summary

Logic: Definition & Purpose - Propositional Logic

Proposition: Atomic & Compound Proposition, Propositional Variable.

Logical Operators: Negation, Conjunction, Disjunction, Exclusive OR, Implication - Converse, Inverse, Contrapositive - Biconditional - Precedence of operators.

Translating English Sentences to Logical Statements.

Tautology, Contradiction, Contingency, Valid, Invalid, Satisfiability, Unsatisfiability.

Logical Equivalences: Identity, Domination, Idempotent, Double Negation, Commutative, Associative, Distributive, De Morgan's, Absorption, & Negation Laws, Conditional & Biconditional Equivalences - Logical Implication - Applications.

Rules of Inference in Propositional Logic - Premises, Conclusion, Argument.

Types of Inference Rules: Modus Ponens, Modus Tollens, Hypothetical Syllogism, Disjunctive Syllogism, Addition, Simplification, Conjunction, Resolution.

Building Arguments using Rules of Inference.

Limitations of Propositional Logic.

Summary

Predicate Logic: Predicates & Truth Values.
Quantifiers: Universal & Existential Quantifiers.
Expressing Quantifiers in English - Quantifiers with Restricted Domain.
Common Quantifier Meanings - Predicates vs Quantifiers - Applications.
Logical Equivalences involving Predicates and Quantifiers.
Universal & Existential Quantifiers - Negating Quantified Expressions.
Nested Quantifiers - Negating Nested Quantifiers.Translating English Sentences.
Resolution Principle: Literal, Clause, Resolvant.
Theorem: Resolvant vs Logical Consequence - Proving Argument Valid or Not.
Fallacies: Fallacy of Affirming Conclusion, Fallacy of Denying Hypothesis.
Rules of Inference for Quantified Statements: Universal & Existential Instantiation & Generalization, Universal Modus Ponens Rule, Universal Modus Tollens Rule.
Normal Forms: Minterm, Maxterm - DNF, CNF - Principal Normal Forms: PDNF, PCNF - Conversions - Properties - Propositional Logic vs Predicate Logic.

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